Repetitive Band Sampling for Finding Endmembers in Hyperspectral Imagery

Chein-I Chang, *Life Fellow, IEEE*, Lulu Liu, Meiping Song, Chunyan Yu, *Senior Member, IEEE*, Tingting Yang, and Hongju Cao

***Abstract*—Band sampling (BSam) is a new concept recently developed for hyperspectral imaging (HSI). It implements a two-stage process by randomly sampling a partial band subset from full bands in the first stage, followed by a series of feedback loops in the second stage, which captures spatial information from the processed data to augment the current data cube in an iterative manner. Its potential and promise have been demonstrated in hyperspectral anomaly detection and image classification. However, a direct application of BSam to endmember finding (EF) is not feasible because endmembers are considered as pure signatures, and using spatial filters not only smears but also contaminates the signature purity of endmembers. To address this issue, this article rederives BSam as a new BSam without feedback, called repetitive BSam (RBSam), which utilizes a repetitive process rather than an iterative feedback process to make BSam applicable to EF. In particular, the endmembers found by an EF algorithm are not necessarily true endmembers, in which case, such EF algorithm-found endmembers are referred to as virtual endmembers (VEs). Since virtual dimensionality (VD) is used to determine the number of endmembers, n_e is only an estimate and may not be accurate in real, practical applications. To resolve this dilemma, a progressive RBSam (PRBSam) is further extended for this purpose. Extensive experiments show that PRBSam not only can perform well on EF but also can actually approximate the true value of n_e as close as possible.**

***Index Terms*—Band sampling (BSam), progressive RBSam (PRBSam), repetitive BSam (RBSam), virtual dimensionality (VD).**

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Chein-I Chang is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, also with the Department of Electrical Engineering, National Cheng Kung University, Tainan 70101, Taiwan, and also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County (UMBC), Baltimore, MD 21250 USA (e-mail: cchang@umbc.edu).

Lulu Liu, Meiping Song, and Chunyan Yu are with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China (e-mail: lll_12262022@163.com; 49322784@qq.com; 36202986@qq.com).

Tingting Yang is with the College of Information Science and Technology, Bohai University, Jinzhou 121013, China (e-mail: yangtingting@bhu.edu.cn).

Hongju Cao is with the School of Software, Dalian University of Foreign Languages, Dalian 116044, China (e-mail: hongjucao1980@sina.com).

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I. Introduction

An ENDMEMBER is defined as an idealistic pure spectral signature [1]. Due to the low spectral resolution of a multispectral image, endmembers generally do not exist, and thus, finding endmembers is not of major concern in multispectral imaging (MSI). However, this is not true for hyperspectral imaging (HSI), which has a very fine spectral resolution that can uncover many unknown material substances but cannot be identified by MSI. Endmembers are such substances and happen to be crucial elements for HSI analysis. So, finding endmembers is one of the primary tasks in hyperspectral data exploitation because an endmember can be used to specify a spectral class and provides critical information in data analysis. For example, in spectral unmixing (SU), endmembers can be used as basic material constituents to represent a hyperspectral image via linear mixtures of these endmembers [2]. For mixed pixel classification, endmembers are material substances that are mixed to form a single pixel for classification [3]. From a data representation point of view, endmembers also play a role similar to eigenvectors in principal component analysis (PCA), where eigenvectors produce principal components to form linear mixtures of eigenvectors to represent a low rank subspace for the data, so as to achieve data dimensionality reduction (DR). A key difference between eigenvectors and endmembers lies in the fact that eigenvectors are generated by a data covariance matrix, and they are not original data samples. By contrast, endmembers are true data samples that can actually be found directly from the data. That is why endmembers can be used for spectral unmixing and mixed pixel classification, while PCA cannot. Nevertheless, both endmembers and eigenvectors can achieve effective DR for hyperspectral image representation.

In order for endmembers to faithfully represent the data, two issues must be addressed, i.e., how to determine the number of endmembers to be present in the data, n_e, and how to find n_e desired endmembers from the data.

As for the first issue, n_e is generally unknown and relatively small compared to full band dimensionality, L, i.e., $n_e \ll L$. Because each endmember is a pure signature, it indicates that only one single spectral component can accommodate one specific endmember in the endmember-unmixed data. It results in a hyperspectral data cube that only requires n_e spectral dimensions to represent the data instead of using full band dimensionality, L.

Alternatively, we can also interpret endmembers in terms of the pigeon-hole principle [4]. Suppose that spectral bands

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and endmembers are pigeon-holes and pigeons, respectively. In HSI, the number of spectral bands, L, is generally greater than the number of endmembers, i.e., $n_e < L$. Thus, this implies that there are more pigeon-holes to accommodate fewer pigeons with $n_e < L$. According to the pigeon-hole principle, more than one pigeon-hole (i.e., more than one spectral band) can accommodate one pigeon (i.e., one endmember). In order to ensure that one pigeon can fly into only one pigeon-hole, the number of pigeon-holes must be reduced to the number of pigeons. In doing so, the orthogonality principle (OP) [4] can accomplish this goal. This explains exactly why OP is used to reduce L to n_e.

To address the above two issues, two concepts of endmember finding (EF) and virtual endmembers (VEs) are introduced in this article. As for the first issue, a notion of virtual dimensionality (VD) was coined and envisioned by Chang [5] to estimate the number of spectrally distinct signatures present in a hyperspectral data, its value denoted by n_{VD}, which can then be used to approximate n_e. Since endmembers are also spectrally distinct signatures, n_{VD} is at least equal to or greater than n_e, i.e., $n_{VD} \geq n_e$.

Regarding the second issue, one must clarify what the so-called EF is. Over the past years, the term endmember extraction (EE) has been widely used for finding endmembers. Technically speaking, EE and EF are completely different concepts. According to [6], EE is a technique that extracts endmembers from the data, where the underlying assumption imposed on EE is that the endmembers must be known to exist in the data a priori. In this case, EE is performed in an active mode in the sense of searching for the known endmembers in the data. To the contrary, EF is a technique that finds endmembers from the data without assuming the existence of endmembers in the data. In this case, EF is performed in a passive mode, which intends to find potential endmember candidates in the data, called VEs in this article. So, if there are no true endmembers present in the data, i.e., $n_e = 0$, the endmembers found by an EF algorithm will be VEs, in which case EF is simply a technique to make an attempt to find VEs as endmember-like signatures, where the number of VEs, n_{VE}, can be determined by n_{VD}. Such EF algorithm-found endmembers are supposed to be very close or to approximate true endmembers. On the other hand, when there are true endmembers present in the data, EF is reduced to EE, which extracts VEs as true endmembers. This follows a relationship of $n_e \leq n_{VE} \leq n_{VD}$. Using this interpretation, many EE algorithms reported in the literature are indeed EF algorithms, for example, the automatic target generation process (ATGP) extended from unsupervised OSP [7], vertex component analysis (VCA) [8], simplex growing algorithm (SGA) [9], etc. In particular, the well-known N-finder algorithm (N-FINDR) developed by Winter in [10] is actually an EF algorithm, as it was named as N-FINDR, not to be called N-extractor, where N is the number of endmembers to be found.

A. Motivation

In EE/EF, the most challenging issue is to determine the true value of n_e. However, without ground truth, validating the true value of n_e is impossible. So, most works reported on EE in the literature did not really address this issue, but rather determined n_e either empirically or by visual inspection. Unfortunately, an incorrect n_e generally has a significant impact on hyperspectral data representation, specifically, SU. As noted above, n_{VD} provides a practical means of estimating n_e, which is usually overestimated because an endmember is indeed a spectrally distinct signature but not vice versa. So, n_{VD} can be used as an upper bound on n_e for EE and on n_{VE} for EF. Once n_e or n_{VE} is determined, a follow-up issue is how to find n_e endmembers by an EE algorithm or n_{VE} VEs by an EF algorithm. Apparently, very little work on this issue has been conducted in the past.

B. Proposed Approach

According to the pigeon-hole principle in [4], each endmember requires a spectral dimension for its accommodation. Consequently, only n_e spectral dimensions are needed for EE, and n_{VE} spectral dimensions are needed for EF. In either case, DR is highly desired as a preprocessing step with $n_{DR} = n_e$ for EE or $n_{DR} = n_{VE}$ for EF. To retain the integrity of each endmember or VE, band selection (BSel) is generally preferred for DR, in which case n_{DR} equals the number of selected bands, n_{Bsel}.

Recently, a new concept of band sampling (BSam) was developed by Chang [11] with its root in compressive sensing (CS) [12], [13]. It replaces random signal sampling (RSS) used by CS with random BSam (RBS) via BSam rate (BSamR), n_{BSam} no greater than n_{DR}. That is, BSam can achieve DR with dimensionality reduced to n_{BSam}, which is equal to or lower than n_{DR}. More specifically, BSam only needs n_{BSam} bands, which may actually be less than n_{DR}. Its success was demonstrated in anomaly detection [14] and hyperspectral image classification [15]. However, the key difference between BSel and BSam is that the former requires a specific BSel strategy, such as band prioritization or a BSel algorithm to select an appropriate band subset, while the latter does not select bands but rather randomly samples bands.

It seems natural and straightforward to apply BSam to EF. Unfortunately, this is not feasible because a VE map (VEMap) is a binary map that only shows the spatial locations of found VEs. The spatial filters implemented in BSam for the feedback process produce real-valued maps that actually smear and contaminate the VEMap. So, the utilization of spatial filters to capture spatial information of a VEMap does not work.

Since BSam takes advantage of CS to replace RSS with RBS, it can achieve the number of bands to be sampled, n_{BSam}, smaller than that required by traditional BSel methods. However, this is traded off against its use of a random strategy, which results in inconsistent sampled bands. Interestingly, through a feedback process, BSam can reduce this randomness. Unfortunately, the feedback process used in BSam no longer works for EF. This is because the feedback process makes use of spatial filters to capture spatial correlation from the processed data, which destroys the signature purity of the found endmembers. An alternative approach is to use a feedforward network. However, it does not work either because

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it also uses local windows, such as receptive fields, which smooth the signature purity of found endmembers.

To avoid using a feedback or feedforward process, which ruins the signature purity of an endmember, a novel approach is to use a so-called repetitive process, which repeats the same RBS process independently over and over again. More specifically, a repetitive process is developed to repeatedly implement RBS independently so that the results are obtained via integrating the results from the previous repetitions of RBS with the results produced by the current repetition by taking their unions. If two consecutive repetitions produced the same results, then the repetitive process is terminated. Such a resulting process is called repetitive BSam (RBSam), which is a new approach of using BSam without using feedback, where n_{RBSam} is defined as the number of bands sampled by RBSam.

In order to have a better estimate of n_{RBSam}, RBSam is further extended to a progressive RBSam (PRBSam). Its idea is to begin with a lower value of n_{RBSam}, smaller than n_{VD}, as an initial estimate and then gradually increase n_{RBSam} by one at a time progressively until it satisfies a stopping rule or it reaches the upper bound n_{VD}. Such PRBSam takes advantage of RBSam using n_{RBSam} lower than n_{VD} to find approximate n_{e} or n_{VE}.

C. Contributions

There are four contributions summarized as follows.

1) A new approach to BSam, called RBSam, is derived for EF. It is neither a feedback process nor a feedforward process, but rather a repetitive process.
2) An extension of RBSam to a progressive version, called PRBSam, is also derived. It allows RBSam to start with a lower n_{RBSam} and gradually improve the RBSam performance.
3) This article introduces EF and the notion of VEs, and further clarifies the issues of EE versus EF and true endmember versus VE.
4) To demonstrate that n_{RBSam} can be lower than n_{VD}, extensive experiments are performed, and a comprehensive study and analysis are also conducted.

The organization of this article is summarized as follows. Section II addresses the issues arising in EF reported in the literature. Section III briefly reviews the idea of BSam and explains why BSam cannot be directly applied to EF, and then further develops a new approach to EF, RBSam for EF. Section IV extends RBSam to PRBSam for EE/EF to deal with an inaccurate estimate of n_{e}. Section V clarifies and compares the differences between RBSam and BSam. Section VI discusses images to be used for experiments and also EF algorithms to be compared to validate the advantages of RBSam, PRBSam, and RPBSam. Section VII conducted detailed synthetic experiments. Section VIII presents real image experiments and discussions. Finally, Section IX summarizes the main results derived from this article.

II. Related Works to EF

Over the past years, many works developed and reported in the literature claimed to be EE but actually EF. Most of them were focused on developing EF algorithms with several critical issues overlooked.

First of all, in real-world application scenarios, there is no prior knowledge of ground truth available to know the true value of n_{e}, which must be determined beforehand, prior to implementing an EF algorithm. This n_{e} indeed determines data dimensionality to be retained and has a crucial impact on the results produced by EF algorithms. In the past, VD [5], [16], [17], [18] and hyperspectral signal identification by minimum error (HySime) [19] have been used for this purpose. However, their estimated values for n_{e} are generally not accurate. For a dataset, HySime estimated a fixed value for n_{e} regardless of applications, but VD-estimated n_{e} can vary with false alarm probability according to different applications. As a consequence, VD has been more flexible and reliable than HySime, as shown in [20] and [21].

Once n_{e} is determined, a second issue is to implement a DR technique to reduce the full data dimensionality of L to the data dimensionality of n_{e}. Since each endmember is a pure signature and is supposed to be accommodated by a single spectral band. Preserving band integrity is crucial. Finding an appropriate DR technique is also critical.

A third issue is the criterion used to find endmembers by an EF algorithm. Due to the purity of endmember signatures, it must satisfy two abundance constraints, the abundance sum-to-one constraint (ASC) and the abundance non-negativity constraint (ANC) [4]. One of the most widely used full (ASC, ANC) constraints criterion is simplex volume (SV). The most celebrated EF algorithm is the maximum volume-based method, N-FINDR, developed by Winter [10] and then followed by VCA in [8] and SGA [9]. In this case, n_{e} determines how many vertices of a simplex are required and also how many dimensions are needed to be retained for DR to calculate an SV. Depending upon how the two abundance constraints ASC and ANC are imposed on EF, various EF algorithms were developed [6], [18], for example, abundance unconstrained EF algorithms, such as the UOSP method, ATGP, ANC-constrained methods such as VCA, and (ASC, ANC)-constrained methods such as N-FINDR and SGA. A detailed comparative study and analysis among ATGP, VCA, and SGA can be found in [25].

Interestingly, a recently developed concept, BSam, seems to resolve all the above-mentioned issues. It can be considered as a dynamic BSel, which randomly samples bands as opposed to the traditional BSel, which can be considered as a static BSel to select a fixed set of bands. As a result, BSam can dynamically find an appropriate band set to accommodate desired endmembers. In particular, BSam allows an EF algorithm to use BSamR, i.e., the number of bands to be sampled, n_{BSam}, smaller than the VD-estimated value, n_{VD}, to effectively deal with the issue of unknown n_{e}. Unfortunately, BSam was originally developed for hyperspectral detection and classification by taking advantage of spatial information captured by a set of spatial filters. Apparently, such an approach does not work for EF, as noted in the introduction. Accordingly, a new look into BSam must be sought.

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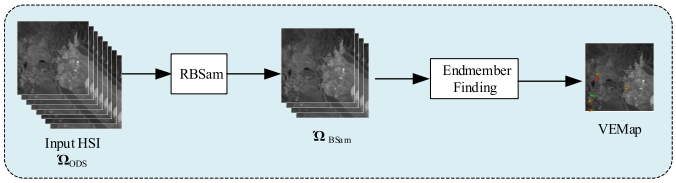


Fig. 1. Graphic diagram of implementing RBSam at the jth progression.

III. Repetitive Random BSam for EF

A. Band Sampling

As originally designed, BSam is derived by replacing RSS used in compressive sampling/sensing [12], [13] with RBS. It involves three key elements. One is to use RBS to randomly sample a partial subset of bands from a full band set according to BSamR, $n_{\text{BSam}} = m$. Another is to use a set of spatial filters to capture spatial information from the processed image data, such as detection maps produced by a detector, classification maps produced by a classifier. A third one is an iterative process that feeds the spatial filter-captured spatial information back to augment the current data cube as a new dataset for the next round of iteration. The system performance is then improved throughout an iteration process carried out via feedback loops by randomly resampling bands repeatedly for each iteration.

Unfortunately, when BSam comes to EF, an EF algorithm-produced VEMap is not a real-valued map but rather a binary map which only shows the spatial locations of found VEs. So, the spatial filters implemented by BSam will actually smooth rather than strengthen their spatial locations. So, this approach is not workable. This implies that a direct application of BSam to EF is not feasible. In other words, the advantage of using spatial filters to capture spatial information via feedback to improve system performance becomes a disadvantage for EF since a VEMap locates exactly the spatial positions of the found VEs, which cannot be compromised by their signature purity. So, the following BSam algorithm is implemented for EE/EF without using a feedback process, and its graphic block diagram of implementing BSam without feedback for EF is depicted in Fig. 1.

B. Repetitive BSam

The primary reason that BSam works well is mainly due to its ability to resolve mixed pixel issues arising in hyperspectral detection and classification, such as boundaries. However, endmembers are completely opposite to mixed pixels, where in EF the integrity of signature purity must be preserved and retained during data processing, and there is no boundary issue. So, in order to still take advantage of the key idea of BSam, we must look into how to implement BSam without using a feedback process as described by Algorithm 1 in Fig. 1.

Due to the nature of randomness caused by the BSam-EE/EF implemented in Fig. 1, each time, the VEMap will be different. In this case, we can convert this disadvantage to an advantage by repeatedly implementing BSam-EE/EF as many times as needed, where after each repetition, the resulting VEMap takes the union of all VEMaps produced by previous repetition with the VEMap produced by the current repetition. To terminate this repetitive process, if the VEMaps produced by two consecutive runs of RBS are identical, RBS is terminated, in which case the found endmembers form the final set of endmembers. Repeatedly implementing BSam without feedback is referred to as RBSam, which replaces an iterative feedback process originally designed for BSam with a repetitive process without using feedback.

Algorithm 1 *BSam Without Feedback for EE/EF*

Input: HSI data $\mathbf{X}$, band sampling rate n_{BSam}
Output: VEndmember Map (VEMap)
Initialize: Let full spectral band set be $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$.
Random Sampling: Randomly select n_{BSam} bands from $\mathbf{\Omega}$ to form a subset $\mathbf{\Omega}_{\text{BSam}} = \{\mathbf{B}_l\}_{l=1}^{n_{\text{BSam}}}$.
Execution: Apply EF algorithm to the data cube restricted to bands in $\mathbf{\Omega}_{\text{BSam}}$.
Output: Generate a binary VEMap representing spatial locations of found endmembers.

The idea of developing RBSam is inspired by that used to design random N-FINDR (RN-FINDR) in [18] and [26]. Due to the use of random data sampling to generate initial conditions, a different run of N-FINDR yields a different set of endmembers. To resolve this inconsistency issue, RN-FINDR is particularly designed to run N-FINDR using different sets of initial conditions, which are generated by randomly sampling data samples, independently, as a repetitive process. Since the endmembers found by N-FINDR are VEs, some of them may not be true endmembers. So, RN-FINDR takes the intersection of VEs found by N-FINDR in different runs as a stopping rule. When two consecutive runs of N-FINDR yield the same set of VEs, RN-FINDR is terminated.

Now, if we replace random data sampling with RBS, the repetitive process used by RN-FINDR is readily applied to BSam without feedback for EE/EF in Fig. 1 to derive RBSam. However, there is a niche between RN-FINDR and RBSam. That is, due to using random band subsets by RBS in independent runs, the set of VEs produced by an EF algorithm may not contain all true endmembers. So, to address this issue, we replace the intersection of VEs used in RN-FINDR with the union of VEs found by an EF algorithm implemented in different runs by RBS. Then, this resulting union is used as a stopping rule to terminate RBSam.

It is worth noting that using the union as a stopping rule seems natural and intuitive once it is proposed. In fact, it does not come easy. This union criterion is based on two unique properties of endmembers. One is the spatial locations of endmembers which cannot be evaluated by many commonly used criteria such as signal-to-noise (SNR), root-mean-squared error (RMSE), etc. The other is the number of endmembers, which is generally very small. So, taking a union is a logical approach. Specifically, an endmember is a spectrally distinct signature. An EF algorithm designed for finding VEs is supposed to be able to find the most probable endmembers, if not all endmembers. This implies that no matter how bands are resampled, an EF algorithm should be able to find most probable endmembers in different runs. In this case, taking the

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Algorithm 2 *RBSam for EE/EF*

Input: Full band set $\mathbf{\Omega}$, n_{BSam}, stopping parameters n_{GT} or n_{term}.
Output: Final integrated endmember set VE_{final}
Initialize: $k \leftarrow 0$; $VE_{total} \leftarrow \emptyset$.
Repeat: Randomly sample $\mathbf{\Omega}^k_{\text{RBSam}} = \left\{\mathbf{B}^k_{l_j}\right\}_{j=1}^{n_{\text{RBSam}}}$ from Ω.
Find k^{th} endmember set $VE^{(k)}$ using an EF algorithm on $\mathbf{\Omega}^k_{\text{RBSam}}$.
Union Operation: Update cumulative set $VE_{total} \leftarrow VE_{total} \cup VE^{(k)}$a.
$n_k = |VE_{total}|$
Check Stopping Rules:
Rule a: If $n_k = n_{\text{GT}}$, **break**.
Rule b: If VE_{total} remains unchanged for two consecutive runs, **break**.
Rule c: If $n_k = n_{term} = L$, **break**.
$k \leftarrow k + 1$.
Until termination condition is met.
Return $VE_{final} = VE_{total}$.

Algorithm 3 *PRBSam for EE/EF*

Input: Initial value $m^{(0)}$, p, data $\mathbf{X}$.
Output: finalVEMap
Initialization: Set $j \leftarrow 0$; determine initial p and $m^{(0)}$ by n_{VD}.
While $n_j < p$ and $m^{(j)} \leq L$ **Do**
$m^{(j)} \leftarrow m^{(0)} + j$
Do $k \leftarrow 0; VE^{(j)} \leftarrow \emptyset$
Repeat:
Implement RBSam with rate $m^{(j)}$.
Update progressive endmember set $VE^{(j)}$ by union of results.
$k \leftarrow k + 1$
Untill $VE^{(j)}$ satisfies the stopping rule defined in Algorithm 2.
$n_{j+1} \leftarrow \left|VE^{(j)}\right|, j \leftarrow j + 1$
End While
Return Final VEMap based on $VE^{(j)}$

union of endmembers found by each RBS should include all possible endmembers if the number of times running BBS is sufficiently large. In order to come up with a stopping rule, three scenarios are of interest. Assume that n_k is the number of VEs in the kth VE set produced by the kth repetition. One is to set n_k to n_{GT}, which is the number of true endmembers that is known. A second one is $n_{k-1} = n_k$. A third one is to set $n_k = n_{\text{term}}$, which is the upper bound on the number of repetitions required to run RBSam.

The detailed implementation of RBSam is described in Algorithm 2.

One comment on n_{term} is noteworthy. Due to the fact that the found VE sets oscillate with only a very few VEs different from two consecutive repetitions, RBS does cause instability. To address this issue, one is to limit the number of repetitions required by RBSam, n_{term}, for example, 20. Another is that n_{term} can be determined by the total number of bands, L. This is because each endmember is supposed to be accommodated by one single spectral dimension. Therefore, the size of the VE set generated by a final repetition cannot go beyond L, which will be imposed as the ultimate upper bound on n_{term}.

Three benefits can be gained from RBSam.

1) It should be noted that the EE/EF algorithm implemented in RBSam is not particularly specified and can be any arbitrary EE/EF algorithm.
2) As a result of repeatedly sampling bands in a random manner by RBSam, each repetition of BSam indeed uses different partial band subsets to perform EE/EF. Taking the union of VEs produced by previous repetitions with VEs produced by the current repetition is actually an integration of VEs produced by an EF algorithm using unions of all partial band subsets randomly sampled by the current and all previous repetitions. With this nature in repetition, RBSam can be considered as a dynamic BSel as opposed to commonly used BSel methods as static BSel, which requires specific algorithms to select a fixed band subset during the entire data processing. As a consequence, if RBSam is run in a sufficiently large number of repetitions, RBSam may eventually exhaust all the possible bands in the full band set Ω.
3) Most importantly, RBSam makes use of a repetitive process without an extra process of spatial filters to capture spatial information for the feedback process. As a result, the computational complexity and efficiency of RBSam are largely reduced compared to the originally designed BSam. This advantage provides a significant benefit for hardware design for implementation.

IV. PRBSam for EE/EF

When RBSam is used, $n_{\text{BSam}} = m$ must be known a priori and then is fixed during the entire process. On many occasions, the value of m is not known exactly, and the estimate of m may also not be accurate. To alleviate this dilemma, we develop a progressive version of RBS, called PRBSam, which implements RBS progressively, starting with a lower value of m and then increasing the value of m by one at a time until it reaches a particular stopping rule.

Let n_{PRBSam} be the number of times that implements PRBSam, and $VE^{(j)}$ be a set of endmembers in the VEMap found by an EF algorithm at the jth progression in PBSam, $\mathbf{X}^{(j)}$ with dimensionality of $m^{(j)}$. Then, the detailed step-by-step implementation of PBSam is summarized as follows, where a pair of two loops, (inner loop, outer loop) = (k, j), are indexed by k and j to indicate the inner and outer loops, respectively. A step-by-step implementation is described in Algorithm 3, where m is defined by n_{BSam}, and p is determined by n_{VD}.

Fig. 2(a) and (b) depicts graphic diagrams of RBSam and PRBSam, respectively. All source codes of the algorithms are made available in website.

https://github.com/LuLuLiu1226/PRBSam.git or
https://umbc.atlassian.net/wiki/spaces/rssipl/pages/27885869/10.+Download

V. Comparison Between BSam and RBSam

This section explores and compares the differences between BSam and RBSam, both of which look similar but indeed are quite different in terms of design rationales and concepts.

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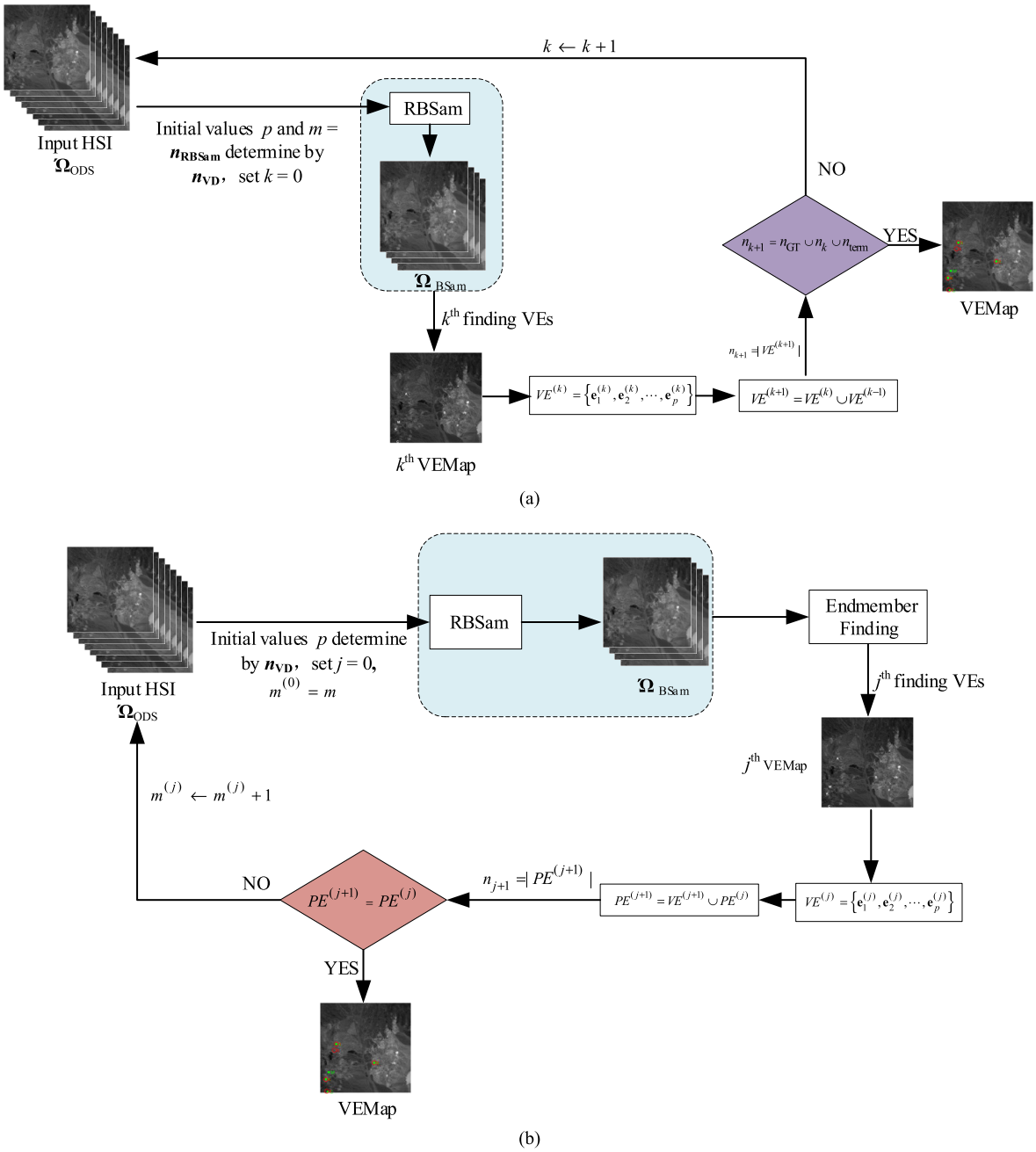


Fig. 2. Graphic diagrams of implementing RBSam and PRBSam. (a) Graphic diagram of implementing RBSam at the kth progression. (b) Graphic diagram of implementing PRBSam.

First of all, BSam is a feedback network system as opposed to RBSam, which is a repetitive network system.

Second of all, BSam is designed for anomaly detection and classification, and utilizes a set of spatial filters to capture spatial information of the processed data (i.e., anomaly detection maps or classification maps), denoted by $\{\text{SFMap}_m^k\}_{m=1}^M$ which are fed back to improve network performance. However, an endmember is a pure signature pixel. Using $\{\text{SFMap}_m^k\}_{m=1}^M$ for feedback will smear and contaminate its spectral signature purity. As a result, RBSam has to discard this approach. Instead, in order to retain the integrity of its found endmembers, RBSam is particularly designed to take unions through a repetitive process rather than a feedback process. The main reason that makes taking union possible for EF is that an endmember is a spectrally distinct target signature, and an algorithm designed for finding endmembers is supposed to be able to find the most probable endmembers, if not all endmembers. This implies that for a given appropriate RBSam rate, no matter how bands are resampled, an EF algorithm should be able to find most endmembers. In this case, taking the union of endmembers found by each RBS should include all possible endmembers if the number of times running RBS is sufficiently large.

Third of all, BSam operates a new data cube which is obtained by taking the union of the with $\{\text{SFMap}_m^k\}_{m=1}^M$ and new randomly–resampled bands, n_{BSam} bands $\mathbf{\Omega}_{\text{BSam}}^k = \{\mathbf{B}_{l_k}^k\}_{l_k=1}^{n_{\text{BSam}}}$ compared to RBSam, which operates only on new

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TABLE I
COMPARISON BETWEEN BSAM AND RBSAM

	BSam	RBSam
Algorithmic structure	Feedback network (iterations)	Repetitive process (repetitions)
Capture spatial information	Implementing a set of spatial filters, $\{\mathrm{SF}_m\}_{m=1}^{M}$ to generate spatial maps at the kth step, denoted by $\{\mathrm{SFMap}_m^k\}_{m=1}^{M}$	No need
Create new data cube at the kth step	Randomly re-sample bands from the original full spectral band set, $\mathbf{\Omega}=\{\mathbf{B}_l\}_{l=1}^{L}$ according to n_{BSam}, denoted by $\mathbf{\Omega}_{\text{BSam}}^k=\{\mathbf{B}_{l_k}^k\}_{l_k=1}^{n_{\text{BSam}}}$ and create new data cube by $\mathbf{\Omega}^k=\bigcup_{i=1}^{k-1}\{\mathrm{SFMap}_m^i\}_{m=1}^{M}\cup\mathbf{\Omega}_{\text{BSam}}^k$	Randomly re-sample n_{BSam} bands from $\mathbf{\Omega}=\{\mathbf{B}_l\}_{l=1}^{L}$, according to n_{RBSam}, denoted by $\mathbf{\Omega}_{\text{RBSam}}^k=\{\mathbf{B}_{l_j}^k\}_{j=1}^{n_{\text{BSam}}}$
Data processing at the kth step	Process data on $\mathbf{\Omega}^k=\bigcup_{i=1}^{k-1}\{\mathrm{SFMap}_m^i\}_{m=1}^{M}\cup\mathbf{\Omega}_{\text{BSam}}^k$ to produced data $A^{(k)}$.	Implement the same EF algorithm on $\mathbf{\Omega}_{\text{RBSam}}^k$ to find the kth endmember set $VE^{(k)}=\{\mathbf{e}_1^{(k)},\mathbf{e}_2^{(k)},\cdots,\mathbf{e}_p^{(k)}\}$ and let $n_k=\lvert VE^{(k)}\rvert$ be the number of endmembers in $VE^{(k)}$ and take union $VE^{(k+1)}=VE^{(k)}\cup VE^{(k-1)}$
Stopping rule	If the overlapped data produced by two successive feedbacks > a given error threshold ε $\mathrm{TI}^{(k)}=\frac{\lvert A^{(k-1)}\cap A^{(k)}\rvert}{\lvert A^{(k-1)}\cup A^{(k)}\rvert}>\varepsilon$ [27]	If two consecutive endmember sets are equal $n_k=n_{\text{GT}}$, $n_{k+1}=n_k$, $n_k=n_{\text{term}}$
Progressive extension	No	PRBSam

randomly–resampled bands, $\mathbf{\Omega}_{\text{RBSam}}^k=\{\mathbf{B}_{l_j}^k\}_{j=1}^{n_{\text{BSam}}}$ without creating a new data cube.

Finally, BSam uses the Tanimoto index [27] of two successive results as a stopping rule, whereas RBSam takes the union of two consecutive endmember sets as a stopping rule.

Table I summarizes the differences between BSam and RBSam in detail.

It should be noted that RBSam and BSam share the property, that is, they both randomly resample $\mathbf{\Omega}_{\text{BSam}}^k=\{\mathbf{B}_{l_k}^k\}_{l_k=1}^{n_{\text{BSam}}}$ in an iterative manner indexed by k and $\mathbf{\Omega}_{\text{RBSam}}^k=\{\mathbf{B}_{l_j}^k\}_{j=1}^{n_{\text{BSam}}}$ in a repetitive manner indexed by k. This repeated resampling band has a great advantage over the fixed sampling bands in the sense that BSam and RBSam actually use the union of band subsets randomly sampled iteration by iteration and repetition by repetition, respectively. For example, the best scenario is the case where each iteration or repetition resamples completely different bands. In this case, for an L-full band set $\mathbf{\Omega}$, after resampling bands $\lceil L/n_{\text{BSam}}\rceil$ or $\lceil L/n_{\text{RBSam}}\rceil$ times, BSam and RBSam can run through all bands in $\mathbf{\Omega}$ in which case BSam and RBSam in fact use up all bands for data processing. As a result, BSam and RBSam can do significantly better than using only one fixed band subset selected by BSel. Nevertheless, if we run BSam and RBSam a sufficient number of iterations or repetitions, we will eventually exhaust all the bands in $\mathbf{\Omega}$, which is equivalent to using all bands in $\mathbf{\Omega}$.

Despite their shared common property described above, BSam and RBSam indeed differ in how to process the data. More specifically, BSam operates on a new expanded data cube formed by taking the union of $\{\mathrm{SFMap}_m^k\}_{m=1}^{M}$ with, $\mathbf{\Omega}_{\text{BSam}}^k=\{\mathbf{B}_{l_k}^k\}_{l_k=1}^{n_{\text{BSam}}}$, $\mathbf{\Omega}^k=\bigcup_{i=1}^{k-1}\{\mathrm{SFMap}_m^i\}_{m=1}^{M}\cup\mathbf{\Omega}_{\text{BSam}}^k$, whereas RBSam operates on only a new set of resampled bands, $\mathbf{\Omega}_{\text{RBSam}}^k=\{\mathbf{B}_{l_j}^k\}_{j=1}^{n_{\text{BSam}}}$. So, as long as computational complexity is concerned, RBSam is much simpler and easier to implement than BSam.

VI. SYNTHETIC IMAGE EXPERIMENTS

Since real images generally do not have a complete ground truth of the spatial locations of true endmembers to validate EF, this section simulates synthetic images from a real image with complete endmember knowledge [28].

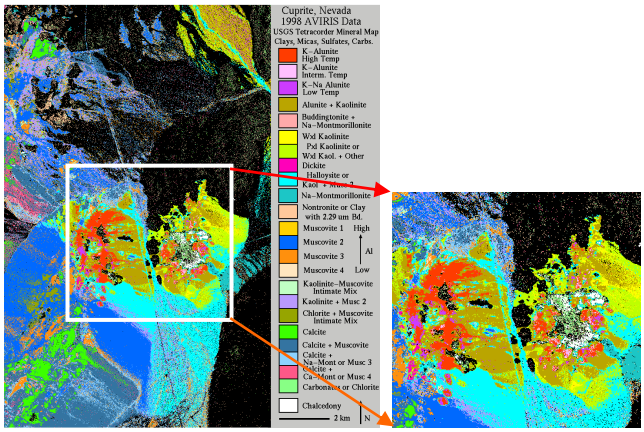
Three EF algorithms were particularly selected and compared for analysis, each of which represents one of three categories: 1) abundance-unconstrained EF algorithm, ATGP; 2) ANC-constrained EF algorithm, VCA; and 3) fully abundance (ASC, ANC)-constrained EF algorithm, SGA.

Let n_{VD} be $p = 6$. In this case, an EF algorithm only needs dimensionality no more than $p = 6$ to find 6 endmembers. This implies that the upper bound of the BSam rate can be set by $n_{\text{BSam}} \le p = 6$. So, a BSel method is used to select p bands to implement ATGP, VCA, and SGA. Despite the many BSel methods that have been proposed in the literature, the self-mutual information-based BSel (SMI-BS) algorithm has been shown to be very effective compared to other existing BSel methods in [29]. So, in the following experiments, SMI-BS and uniform BSel (UBS) were selected and implemented to reduce data dimensionality L to p, along with using full bands for comparison.

It should be noted that VCA was originally designed to use PCA for DR prior to EF. So, for VCA, two versions of VCA were implemented: VCA using PCA and VCA using BSel. Since ATGP and SGA do not require DR, they were implemented by using full bands and BSel.

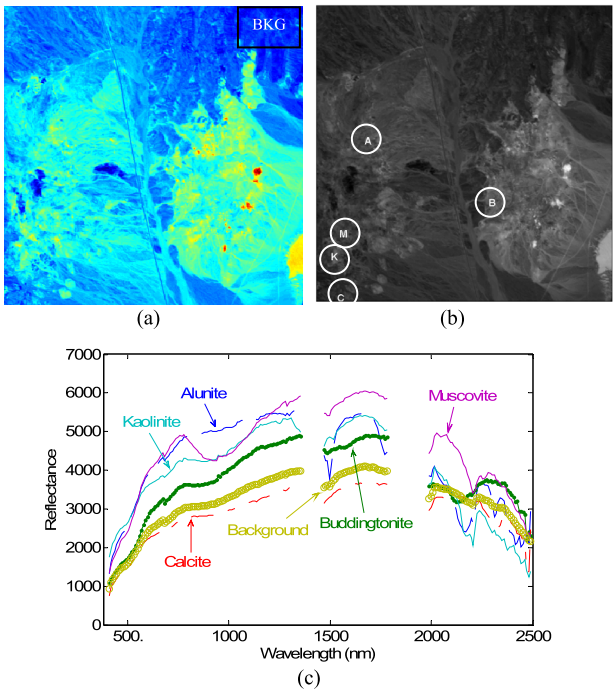
The image data used to simulate synthetic images is shown in Fig. 3(a) and was acquired by the Airborne Infrared Imaging Spectrometer (AVIRIS) over the Cuprite mining site, Nevada,

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(a) (b)

Fig. 3. Cuprite image scene. (a) Original Cuprite image scene. (b) Image cropped from the center region of the original scene in (a) with a size of 350 × 350.



(a) (b) (c)

Fig. 4. (a) Cuprite AVIRIS image scene. (b) Spatial positions of five pure pixels corresponding to minerals: A, B, C, K, M. (c) Five mineral reflectance spectra along with background signature.

in 1997, and used for experiments. It should be noted that bands 1–2 and 221–224 are noisy, and bands 104–113 and 148–167 correspond to water vapor absorption regions; all these bands were removed. As a result, 188 bands remain for analysis in the Cuprite dataset to be used for experiments.

This image scene has 350 × 350 pixels with a spatial resolution of 20 m and 10 nm spectral resolution across the range of 0.4–2.5 μm, collected by 224 bands. This geological scene is the most studied dataset provided by the USGS and is available at website for public use. It is well understood mineralogically and has been used as a standard test site for EF.

An area was particularly cropped from Fig. 3(a) and shown in Fig. 3(b). Fig. 4(a) shows more details of Fig. 3(b) along

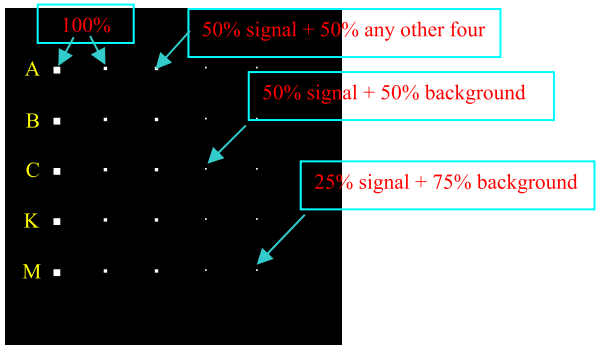


Fig. 5. Set of 25 panels simulated by A, B, C, K, M.

with the ground truth of spatial locations of five pure pixels that can be identified and located to represent five endmembers denoted by alunite (A), buddingtone (B), calcite (C), kaolinite (K), and muscovite (M) in Fig. 4(b) with spectral signatures plotted in Fig. 4(c). Also, the area in the upper right corner of Fig. 4(a) highlighted by BKG in a box was used to simulate the image background signature by calculating the sample mean of the BKG. A detailed study on this scene is also available in [6], [18], and [28].

Fig. 5 shows a synthetic image simulated by 25 panels using five mineral signatures, A, B, C, K, M in Fig. 4(c) according to the legends described in the figure. The details of this synthetic image can be found in [28] and [1, Sec. 4.3]

More specifically, among the simulated 25 panels are five 4 × 4 pure-pixel panels as endmembers for each row in the first column and five 2 × 2 pure-pixel panels as endmembers for each row in the second column, the five 2 × 2-mixed pixel panels for each row in the third column and both the five 1 × 1 subpixel panels for each row in the fourth column and the fifth column where the mixed and subpanel pixels were simulated according to the legends in Fig. 5. So, a total of 100 pure pixels (80 in the first column and 20 in the second column), referred to as endmember pixels were simulated in the data by the five endmembers, A, B, C, K, M. Specifically, an area marked by "BKG" at the upper right corner of Fig. 4(a) was selected to find its sample mean, i.e., the average of all pixel vectors within the area "BKG," denoted by **b** and plotted in Fig. 4(c), to be used to simulate the background for the image scene in Fig. 5. As shown in Fig. 4(c), the spectral signature of **b** is very similar and close to that of Calcite C. It should be noted that the background selection is empirical because the selected area "BKG" seemed more homogeneous than the other regions. Nevertheless, other areas can also be selected for the same purpose. The image background simulated by the signature "**b**" was further corrupted by an additive noise to achieve a certain signal-to-noise ratio (SNR), which was defined as 50% signature (i.e., reflectance/radiance) divided by the standard deviation of the noise in [2].

Once target pixels and background are simulated, two types of target insertion can be designed to simulate experiments for true endmembers to be used for EE and VEs for EF, where VEs are true endmembers corrupted by noise or contaminated by BKG signatures. It is these VEs that are simulated to reflect real-world scenarios when true endmembers may not exist and may not be guaranteed.

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TABLE II
RUNNING TIME IN SECONDS REQUIRED BY ATGP, VCA, AND SGA USING FULL BANDS AND BAND SELECTED BY UBS, SMI-BS, AND RBSAM

	DR		EFAs					
			ATGP		VCA		SGA	
Full band	-		0.1910		-		4.5873	
PCA	0.4339	0.0499	-		0.1329	0.0143	-	
SMI-BS	91.4266		0.0092		0.0123		0.6607	
Uniform-BS	0.0141		0.0076		0.0116		0.6307	
BSam	0.0114	0.0006	0.0074	0.0087	0.0239	0.0097	0.6337	0.6377

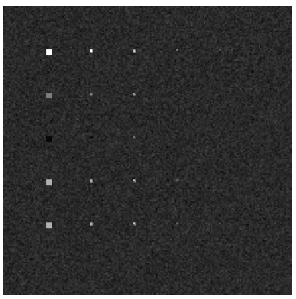


Fig. 6. Synthetic image simulated by scenario TI.

A. Target Implantation

The target insertion simulated in this section is target implantation (TI) which was simulated by directly inserting clean target panels in Fig. 5 into a noisy image background contaminated by additive Gaussian noise with SNR = 20:1. More precisely, the clean targets are implanted in the noisy image BKG by replacing BKG pixels with clean target pixels as shown in Fig. 6. As a result, TI contains true endmembers specified by 100 pure panel pixels located in the first two columns in Fig. 6.

Since TI has 100 true endmembers present in the scene, when EF is applied to TI to find endmembers, EF becomes EE, which extracts true endmembers. According to [6] and [18], the VD of TI estimated by NWHFC is $n_{\mathrm{VD}} = 6$ if $\mathrm{P_F} \leq 10^{-2}$. This estimated n_{VD} is accurate because there are five pure signatures, A, B, C, K, M, plus the BKG signature **b** to represent six spectrally distinct signatures, in which case, $p = 6$, where there are $n_{\mathrm{e}} = 5$ pure signatures, A, B, C, K, M, and one mixed signature, **b**.

1) EE Using Full Bands, BSel, and BSam: Fig. 7 shows endmember maps found by ATGP, VCA, and SGA using full bands, two BSel methods, UBS, SMI-BS, and bands sampled by BSam with $n_{\mathrm{VD}} = n_{\mathrm{BSel}} = n_{\mathrm{BSam}} = 6$ and $n_{\mathrm{e}} = 5$, where the found endmembers are marked by red open circles. Since VCA and BSam used random initial conditions, experiments for two random initial condition scenarios indicated by j were also included to show their inconsistent results. As shown in Fig. 7, only ATGP using UBS and full bands and SGA using full bands could find all five mineral endmembers. Interestingly, the original VCA using PCA developed in [8] could not find any endmember, but VCA used SMI-BS and BSam, and UBS could find three or four endmembers.

The results shown in Fig. 7 indicated that for an EF algorithm to find all the desired five true endmembers, dimensionality reduced to 5 or 6 is not sufficient, except on one special occasion for ATGP using UBS in Fig. 7(a). This is mainly due to spectral signature similarity. For example, among the five mineral signatures, the calcite signature is most similar to **b**, and the muscovite signature is most distinct from **b**. As a result, the extraction of C is most difficult compared to M, which is the easiest to extract, C. In this case, it requires more dimensions to extract. This was confirmed by Fig. 7, where the missing signature was always C, which required using full bands to extract C. Table II also documented their running times, where using BSam yielded the least running time.

2) EE Using RBSam: The endmember results in Fig. 7 show that ATGP, VCA, and SGA did not perform well using BSel and BSam. These facts may lead to the conclusion that an EE algorithm using DR lost critical information for finding endmembers. Fig. 8 shows otherwise. That is, if ATGP, VCA, and SGA are using RBSam with $n_{\mathrm{BSam}} = 6$, they could find all five mineral endmembers successfully, and the value of k underneath each figure indicates the number of repetitions carried out by RBSam with the found endmembers marked by red open circles. Table III also tabulates their running times for each repetition k.

3) EE Using PRBSam: According to the results in Fig. 8, ATGP, VCA, and SGA were able to find all five true endmembers using RBSam with $n_{\mathrm{BSam}} = 6$. An interesting question is "Can we do the same if we lower n_{BSam} less than 6?" PRBSam offers its answer in Table IV, which tabulates $n_{\mathrm{VD}} = p = 6$ endmembers found by ATGP, VCA, and SGA using RBSam with $n_{\mathrm{BSam}} = m$ running from 2 to 5, where the endmembers highlighted in red indicate the new endmembers found at the particular kth repetition, and the endmembers highlighted by blue indicate that these endmembers were still missing and could not be found after PRBSam was terminated. Since too many figures are needed to show the spatial locations of all found endmembers, they are not included to save space. Instead, Table IV should suffice to list all fund endmembers. As can be seen from Table IV, ATGP, VCA, and SGA could find all the five true endmembers using only $n_{\mathrm{BSam}} = m = 2$, 4, and 3, respectively, all of which were lower than $n_{\mathrm{e}} = 5$. These experiments provided evidence that BSam takes advantage of RBS derived from RSS in CS to actually achieve the BSam rate, n_{BSam} less than its required dimensionality, $n_{\mathrm{e}} = 5$.

B. Target Embeddedness

Another target insertion is target embeddedness (TE) which was simulated by embedding (i.e., superimposing) clean target

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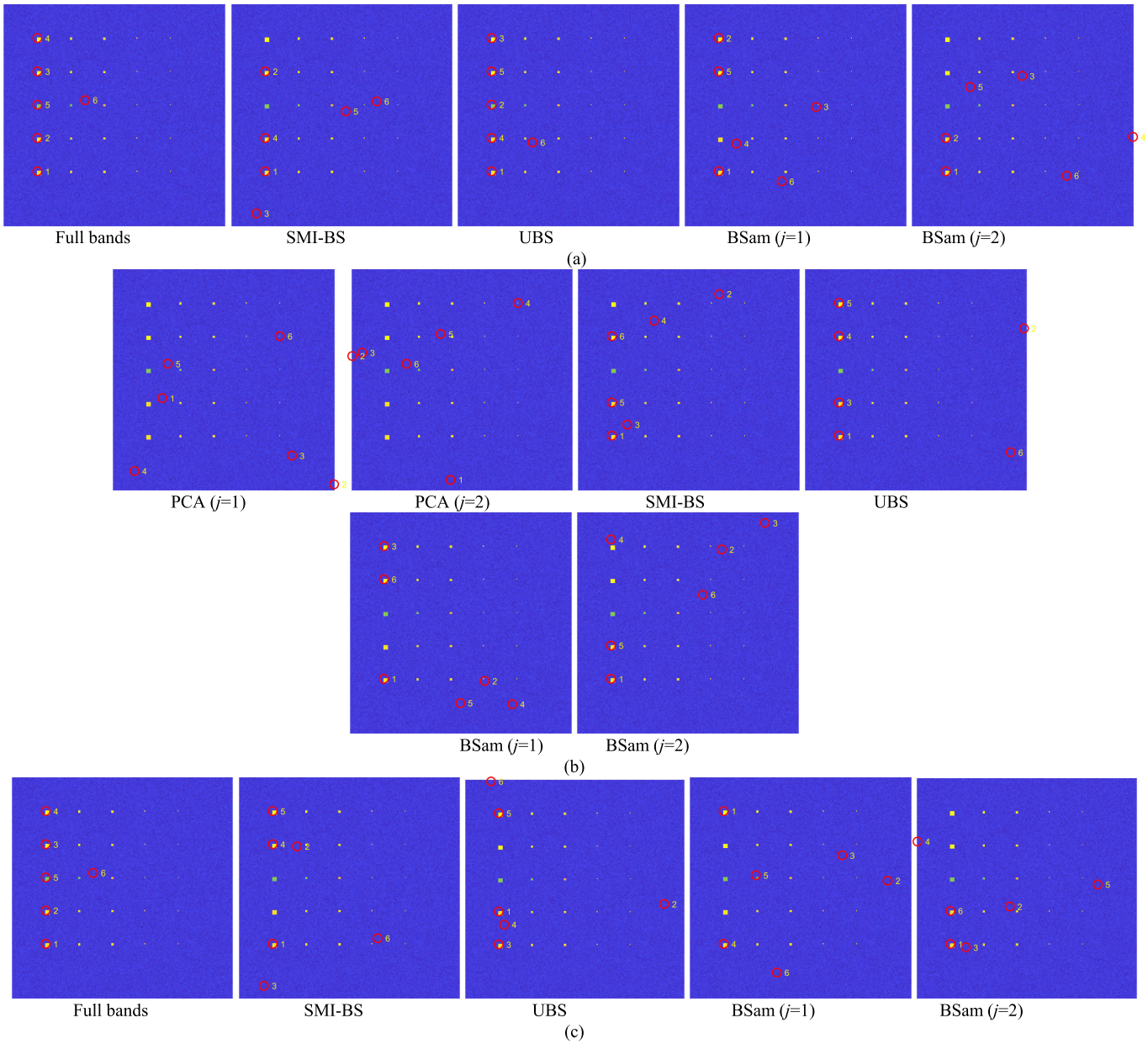


Fig. 7. Endmembers found by (a) ATGP, (b) VCA, and (c) SGA using full bands and bands selected by UBS, SMI-BS, and BSam for TI.

TABLE III

RUNNING TIME IN SECONDS REQUIRED BY ATGP, VCA, AND SGA USING FULL BANDS AND BAND SELECTED BY UBS, SMI-BS, AND RBSAM

	DR	EFAs		
		ATGP	VCA	SGA
k=1	0.5372	0.0117	0.0623	0.6414
k=2	0.0956	0.0098	0.0148	0.6379
k=3	0.0939	-	0.0132	0.6279
k=4	0.1075	-	0.0122	-

panels into the clean image background and then by adding a Gaussian noise with SNR = 20:1 to the clean-target embedded image to yield a Gaussian noise corrupted image, TE as shown in Fig. 9. As a result of TE, there are no true endmembers present in Fig. 9, i.e., $n_e = 0$ because all the 100 clean panel pixels as true endmembers are corrupted by the Gaussian noise. This implies that EE is not applicable to TE because no true endmember can be extracted from TE. This scenario is practical in real-world applications since the presence of true endmembers in a dataset is generally not guaranteed, and endmembers are most likely corrupted by noise or contaminated by BKG signatures. In this case, EF is designed to find the most likely pixels, referred to as VEs, to represent true endmembers.

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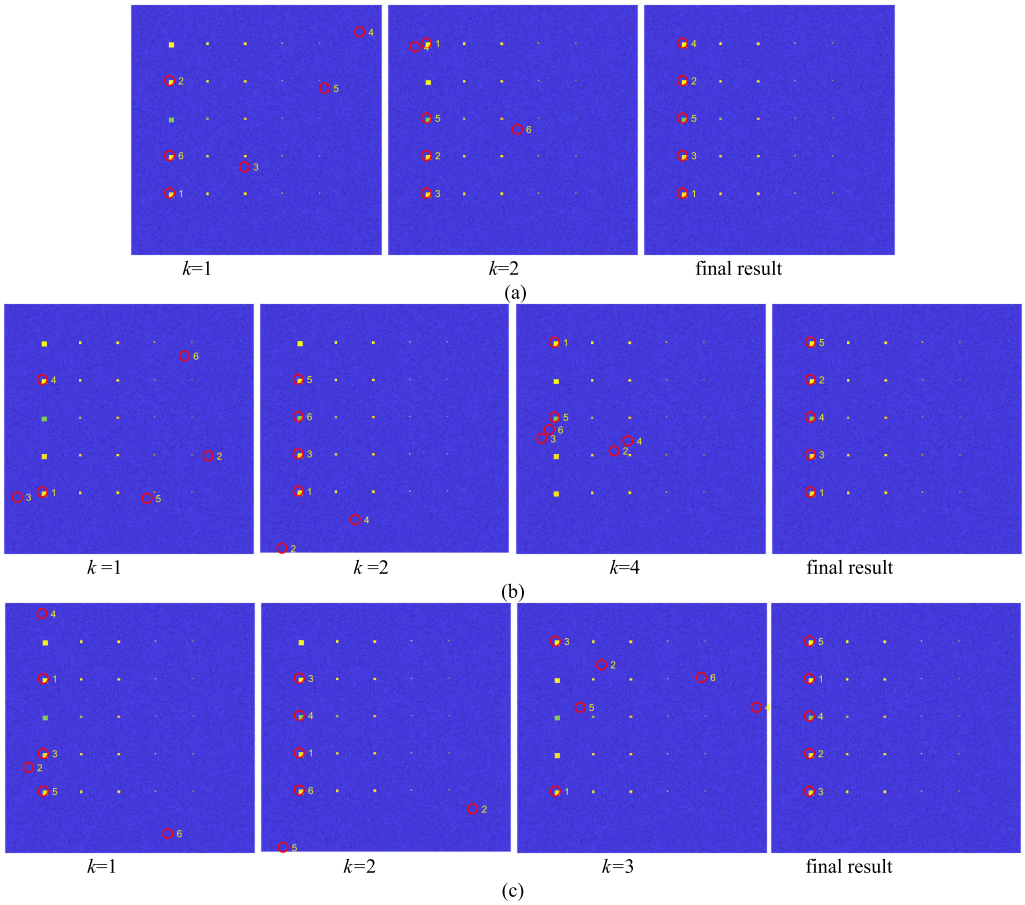


Fig. 8. Endmembers found by (a) ATGP, (b) VCA, and (c) SGA using RBSam for TI.

TABLE IV
PANEL PIXELS FOUND BY ATGP, VCA, AND SGA ON PRBSAM FOR M = 2–6 AND P = 6

Outer loop	Inner loop	ATGP	VCA	SGA
m=2	*k*=1	p_1,p_5	p_5	p_5
	k=2	p_1,p_2,p_5	p_5	p_5
	k=3	p_1,p_2,p_5	p_1, p_5	p_1,p_5
	k=9	p_1,p_2,p_5	p_1,p_2, p_5	p_1,p_5
	k=10	p_1,p_2,p_3,p_5	p_1,p_2, p_5	p_1,p_5
	k=11	p_1,p_2,p_3,p_4,p_5	p_1,p_2, p_5	p_1,p_2,p_5 (missing p_3, p_4)
	k=12	-	p_1,p_2, p_4, p_5 (missing p_3)	-
m=3	*k*=1	p_3, p_5	p_5	p_3, p_5
	k=2	p_3, p_5	p_5	p_3, p_5
	k=3	p_3, p_4, p_5	p_4,p_5	p_3,p_4, p_5
	k=4	p_1, p_3, p_4, p_5	p_1,p_4,p_5	p_1, p_3,p_4, p_5
	k=5	p_1, p_2,p_3, p_4, p_5	p_1,p_4,p_5	p_1, p_3,p_4, p_5
	k=7	-	p_1,p_2,p_4,p_5 (missing p_3)	p_1, p_2,p_3,p_4, p_5
m=4	*k*=1	p_1, p_2, p_4, p_5	p_1, p_5	p_1, p_4, p_5
	k=2	p_1, p_2, p_3, p_4, p_5	p_1, p_4,p_5	p_1, p_4, p_5
	k=3	-	p_1, p_3, p_4,p_5	p_1, p_2, p_4, p_5 (missing p_3)
	k=4	-	p_1, p_3, p_4,p_5	-
	k=5	-	p_1, p_3, p_4,p_5	-
	k=6	-	p_1, p_3, p_4,p_5	-
	k=7	-	p_1, p_2, p_3, p_4,p_5	-
m=5	*k*=1	-	-	p_1, p_3, p_5
	k=2		-	p_1, p_3, p_5
	k=3		-	p_1,p_2, p_4, p_3, p_5

Because there are no true endmembers in TE, a key and crucial issue arises. How many VEs are supposed to be found by an EF algorithm? This number, denoted by n_{VE}, will determine how many VEs can be found in the dataset.

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TABLE V
RUNNING TIME IN SECONDS REQUIRED BY ATGP, VCA, AND SGA USING FULL BANDS AND BAND SELECTED BY UBS, SMI-BS, AND RBSAM

	DR		EFAs					
			ATGP		VCA		SGA	
Full band	-		0.1894		-		4.5056	
PCA	0.0888	0.0446	-		0.1145	0.0120	-	
SMI-BS	52.1810		0.0090		0.0109		0.6542	
Uniform-BS	0.0014		0.0068		0.0108		0.6450	
RBSam	0.0012	0.0005	0.0068	0.0084	0.0231	0.0095	0.6473	0.6470

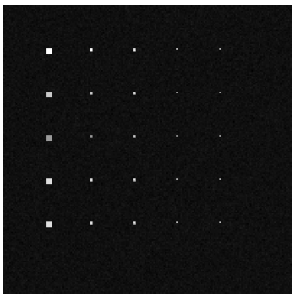


Fig. 9. Synthetic image simulated by scenario TE.

Algorithm 4 *Endmember Identification Algorithm*

1. For each ground truth endmember, m_i, find a found target pixel, denoted by $\mathbf{t}_{j^*}$ in $\{\mathbf{t}_j\}_{j=1}^{J}$ which is closest to m_i by
$$j^* = \arg \min_{1\le j\le p, 1\le j\le J} \{\mathrm{SAM}(\mathbf{t}_j, \mathbf{m}_i)\}$$
2. Cluster all found pixels $\{\mathbf{t}_j\}_{j=1}^{J}$ into class C_i corresponding to the ith endmember m_i via
$$\min_{1\le j\le J} \{\mathrm{SAM}(\mathbf{t}_j, \mathbf{m}_i)\} \Leftrightarrow \mathbf{t}_j \in C_i$$
3. Select the pixel with the smallest SAM value in C_i as the final identified endmember
$$\hat{\mathbf{m}}_i = \arg\left[\min_{1\le j\le J, t_j\in C_i} \{\mathrm{SAM}(\mathbf{t}_j, \mathbf{m}_i)\}\right]$$

Interestingly, the concept of VD was originally designed to estimate the number of spectrally distinct signatures, n_{VD}, in hyperspectral data, which can be used for this purpose. So, n_{VD} is generally greater than or at least equal to n_{VE}, i.e., $n_{\mathrm{VD}} \ge n_{\mathrm{VE}}$. This indicates that an EF algorithm must find at least n_{VD} signatures to guarantee that VEs will be included to be treated as if they were endmembers. Despite the fact that VEs may not be true endmembers, each VE is, in fact, a spectrally distinct signature and can be considered as an endmember-like signature. Now, the follow-up issue is how to identify which found n_{VD} VEs signatures as desired true endmembers. To resolve this dilemma, an endmember identification algorithm (EIA) is developed to identify which signatures from the EF found VEs as desired endmembers. The details of implementing EIA step-by-step are summarized in Algorithm 4.

According to Fig. 9, there are no true endmembers, i.e., $n_{\mathrm{e}} = 0$. Nevertheless, the noise-corrupted mineral signatures are indeed VEs, which turn out to be the purest signatures. The VD of TE estimated by NWHFC is $n_{\mathrm{VD}} = 6$ with $\mathrm{P_F} \le 10^{-2}$. This VD estimate is also accurate since this 6 represents the five spectrally distinct VEs plus a BKG mixed signature **b**. In this case, $n_{\mathrm{VD}} = 6 > n_{\mathrm{VE}} = 5$ and $n_{\mathrm{VE}} = 0$.

1) EF Using Full Bands, BSel, and BSam: Fig. 10 shows VEMaps found by ATGP, VCA, and SGA using full bands, two BSel methods, UBS, SMI-BS, and bands sampled by BSam, where $n_{\mathrm{VD}} = n_{\mathrm{BSel}} = n_{\mathrm{BSam}} = 6$ and $n_{\mathrm{VE}} = 5$, where the found VEs are identified by EIA with SAM <0.05 and marked by red open circles. Since VCA and BSam used random initial conditions, experiments for two random initial condition scenarios indicated by j were also included. Apparently, Fig. 10 shows that the best results could only find four VEs regardless of using full bands or any DR technique. Similar to Fig. 7, no EF algorithms could find all five VEs mainly due to signature variability contaminated by BKG signature **b**. The worst performance was VCA, which could only find two VEs using full bands and BSel, as well as three VEs using BSam. Table V also documented running times to produce endmember results in Fig. 10, where using BSam yielded the least running time.

2) EF Using RBSam: Fig. 10 showed that ATGP, VCA, and SGA did not perform well using BSel and BSam. Interestingly, Fig. 11 shows that the performance of an EF algorithm can be significantly improved by using RBSam, where ATGP, VCA, and SGA using RBSam with $n_{\mathrm{BSam}} = 6$ could successfully identify all five mineral VEs identified by EIA with SAM <0.05, where the final results were obtained by taking unions of results obtained for all k repetitions. The value of k underneath each figure in Fig. 11 indicates the number of repetitions carried out by RBSam, and the found VEs are marked by red open circles. Table VI also tabulates their running times for each repetition k.

3) EF Using PRBSam: As shown in Fig. 11, using RBSam with $n_{\mathrm{BSam}} = 6$, VCA and SGA were able to find all five mineral VEs, and ATGP could find four VEs. Similar to TI, can PRBSam also offer the same advantages for TE? Since too many figures are needed to show the spatial locations of all found endmembers, they are not included to save space. So, only Table VII tabulates $n_{\mathrm{VD}} = p = 6$ VEs found by ATGP, VCA, and SGA using RBSam with $n_{\mathrm{BSam}} = m$ running from 2 to 5, where the VEs highlighted in red indicate the new endmembers found at the particular kth repetition, and VEs highlighted by blue indicate that these VEs were still missing and could not be found after PRBSam was terminated. As can

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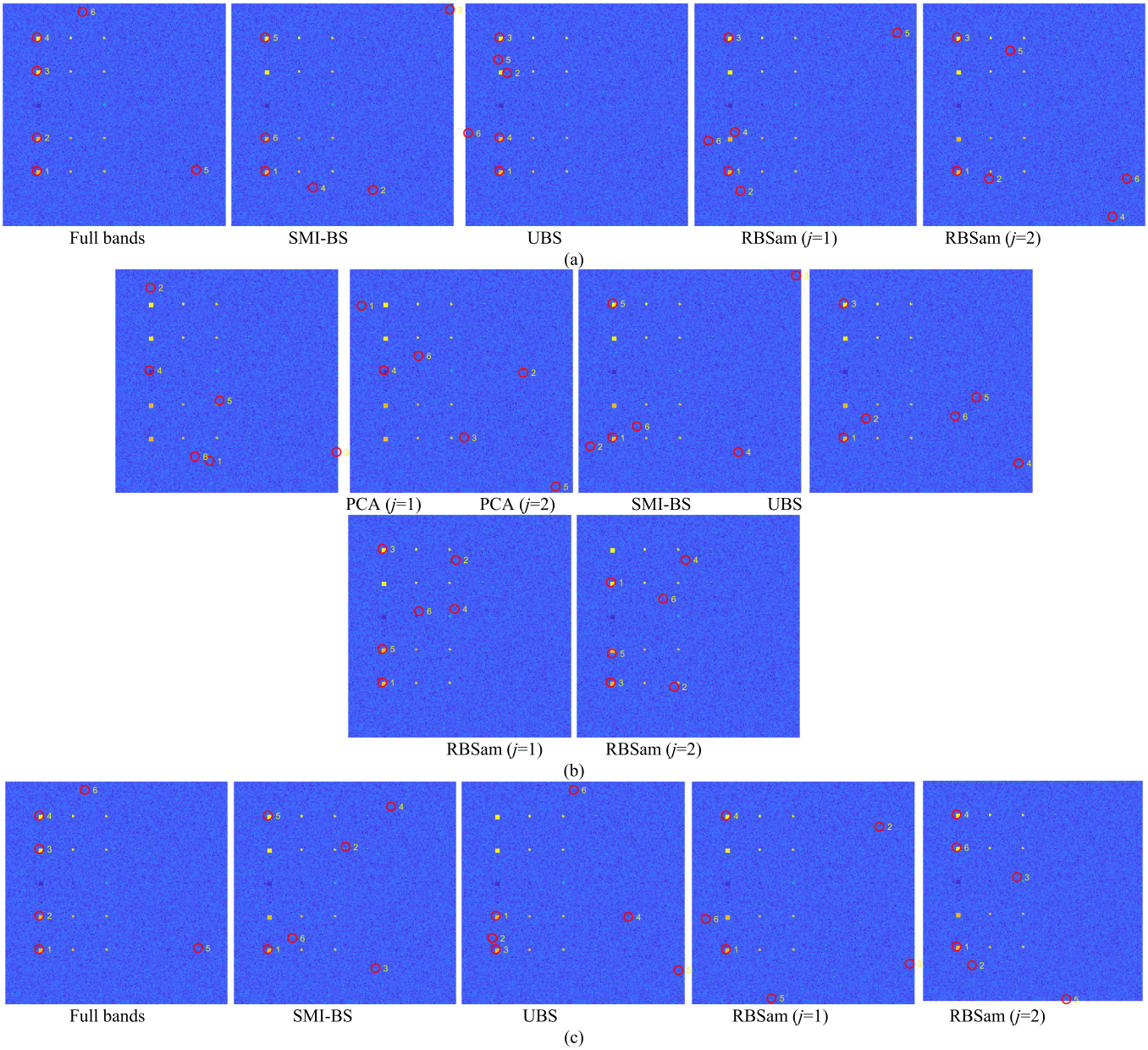


Fig. 10. Endmembers found by (a) ATGP, (b) VCA, and (c) SGA using full bands and bands selected by UBS, SMI-BS, and RBSam for TE.

TABLE VI

RUNNING TIME IN SECONDS REQUIRED BY ATGP, VCA, AND SGA USING FULL BANDS AND BAND SELECTED BY UBS, SMI-BS, AND RBSAM FOR TE

	DR	EFAs		
		ATGP	VCA	SGA
k=1	0.0829	0.0108	0.0570	0.7412
k=2	0.0988	0.0081	0.0118	0.6332
k=3	0.0868	0.0064	0.0110	-
k=4	0.0890	0.0056	-	-

be seen from Table VII, ATGP, VCA, and SGA could find all the five true endmembers using only $n_{\text{BSam}} = m = 5(k = 9)$, $5(k = 5)$, and $3(k = 6)$, respectively, all of which were lower than $n_{\text{VE}} = 6$. These experiments provided evidence that BSam takes advantage of RBS derived from RSS in CS to actually achieve the BSam rate, n_{BSam} less than its required dimensionality, $n_{\text{e}} = 5$.

VII. IMAGES USED FOR EXPERIMENTS

Although many real datasets have been used for SU, such as Cuprite, Jasper Ridge, Samson, Urban [30], [31], [32], [33], [34], [35], most of them are not suitable for EF. This is because EF and SU are completely different tasks. For datasets to be used for EF, the ground truth of spatial locations of EF algorithm-found endmembers or VEs must be provided for

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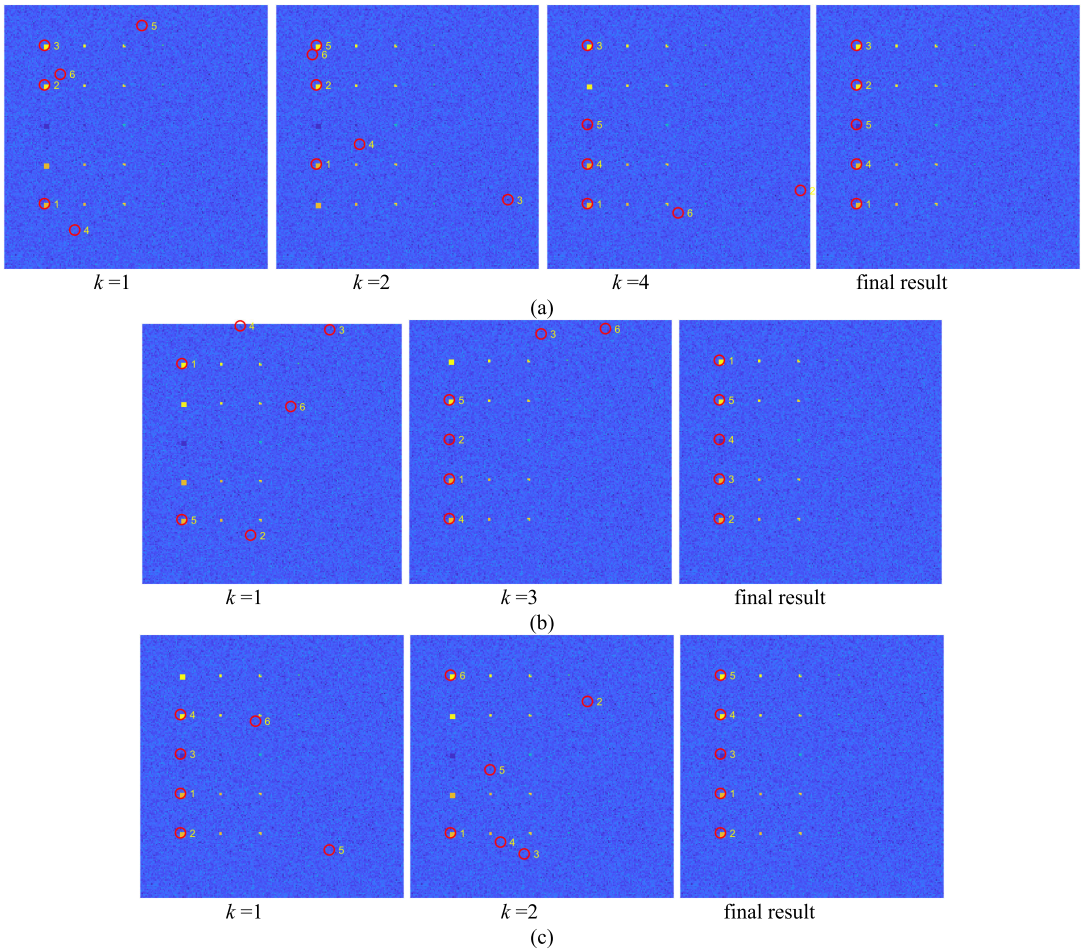


Fig. 11. Endmembers found by (a) ATGP, (b) VCA, and (c) SGA using RBSam for TE.

validation. On the other hand, SU only needs the spectral signatures of endmembers for unmixing, which can be obtained by U.S. Geological Survey with no need to know their spatial locations. Since the datasets used for SU in [30], [31], [32], [33], [34], and [35] do not have a complete ground truth of the spatial locations of the used endmembers, these datasets are not applicable to EF to be used to verify the EF algorithm-found endmembers or VEs. So, in this section, the datasets used for SU in [30], [31], [32], [33], [34], and [35] are not applicable to EF. Instead, we particularly selected two datasets that can be used to validate our findings.

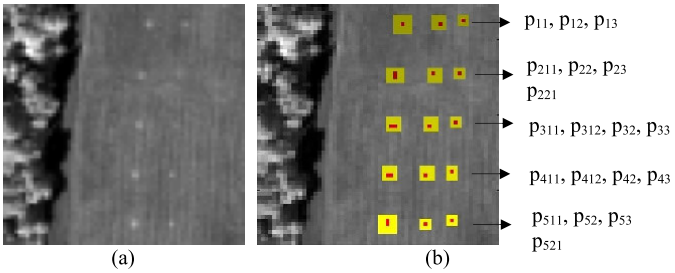


Fig. 12. (a) HYDICE panel scene, which contains 15 panels. (b) Ground truth map of spatial locations of 19 R panel pixels.

A. HYDICE Panel Data

The first real hyperspectral image is an airborne hyperspectral digital imagery collection experiment (HYDICE) scene shown in Fig. 12(a), which was used to analyze comparative performance among various target detectors and anomaly detectors. It was collected in August 1995 from a flight altitude of 10 000 ft. Here are 15 square panels in Fig. 12(b) with three different sizes, 3×3 m, 2×2 m, and 1×1 m, respectively.

Due to the ground sampling distance of approximately 1.56 m, each of the panels in the first column, except the first row, contains two panel pixels highlighted by red, p_{211}, p_{221} in row 2, p_{311}, p_{312} in row 3, p_{411}, p_{412} in row 4, and p_{511}, p_{521} in row 5, as shown in Fig. 12(b). All the remaining 11 panels in Fig. 12(b) contain one single panel pixel for each panel, also highlighted by red, p_{11}, p_{12}, p_{13} in row 1, p_{22}, p_{23} in row 2, p_{32}, p_{33} in row 3, p_{42}, p_{43} in row 4, and p_{52}, p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 3(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. So, according to the provided ground truth, there are a total of 14 true endmembers corresponding to 9 R panel pixels in the first column, p_{11}, p_{211}, p_{221}, p_{311}, p_{312}, p_{411}, p_{412}, p_{511}, p_{521} and five R panel pixels in the second column, p_{12}, p_{22}, p_{32}, p_{42}, p_{52} as shown in Fig. 12 where only five spectrally distinct true endmembers, one for each different row. More

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TABLE VII
PANEL PIXELS FOUND BY ATGP, VCA, AND SGA ON PRBSAM FOR m = 2–6 AND p = 6

Outer loop	Inner loop	ATGP	VCA	SGA
m=2	k=1	p_4, p_5	p_1	p_4, p_5
	k=2	p_1,p_4, p_5	p_1,p_4	p_4, p_5
	k=3	p_1,p_4, p_5	p_1,p_4	p_1,p_4, p_5
	k=4	p_1,p_4, p_5	p_1,p_4	p_1,p_4, p_5
	k=5	p_1,p_4, p_5	p_1, p_4, p_5	p_1,p_4, p_5
	k=6	p_1,p_4, p_5	-	p_1,p_4, p_5
	k=7	p_1,p_2,p_4, p_5	-	p_1,p_4, p_5
	k=8	p_1,p_2,p_4, p_5	-	p_1,p_4, p_5
	k=10	p_1,p_2,p_4, p_5 (missing p_3)	p_1, p_4, p_5 (missing p_2, p_3)	p_1, p_3, p_4, p_5 (missing p_2)
m=3	k=1	p_1	p_2,p_5	p_1, p_5
	k=2	p_1, p_5	p_2, p_4, p_5	p_1, p_4,p_5
	k=3	p_1, p_5	p_1, p_2, p_4, p_5	p_1,p_4, p_5
	k=4	p_1, p_5	p_1, p_2, p_4, p_5	p_1, p_2, p_4, p_5
	k=5	p_1, p_2,p_4, p_5	p_1, p_2, p_4, p_5	
	k=6	p_1,p_2,p_4, p_5	p_1, p_2, p_4, p_5	p_1, p_2, p_3, p_4, p_5
	k=10	p_1,p_2,p_4, p_5 (missing p_3)	p_1, p_2, p_4, p_5 (missing p_3)	-
m=4	k=1	p_1,p_5	p_1,p_4,p_5	-
	k=2	p_1, p_4, p_5	p_1,p_4, p_5	-
	k=3	p_1, p_4, p_5	-	-
	k=4	p_1, p_4, p_5	-	-
	k=5	p_1, p_2,p_4, p_5	-	-
	k=6	p_1, p_2,p_4, p_5	-	-
	k=7	p_1, p_2,p_4, p_5	-	-
	k=8	p_1, p_2,p_4, p_5	-	-
	k=10	p_1, p_2,p_4, p_5 (missing p_3)	p_1,p_2,p_4, p_5 (missing p_3)	-
m=5	k=1	p_4, p_5	p_1,p_2,p_4	-
	k=2	p_1, p_2,p_4, p_5	p_1,p_2,p_4	-
	k=3	p_1, p_2,p_4, p_5	p_1,p_2,p_4,p_5	-
	k=4	p_1, p_2,p_4, p_5	p_1,p_2,p_4,p_5	-
	k=5	p_1, p_2,p_4, p_5	p_1,p_2,p_3,p_4,p_5	-
	k=6	p_1, p_2,p_4, p_5	-	-
	k=9	p_1, p_2,p_3,p_4, p_5	-	-

specifically, the R panel pixels in each row are considered the same type of true endmembers specified by the same spectral signature.

B. Cuprite Data

A second real hyperspectral image is the Cuprite data shown in Fig. 4, where there are five minerals, A, B, C, K, M, with their spatial locations identified by white open circles in Fig. 4(b). Since this scene is a geological scene that has A, B, C, K, M spread over the site, with various spatial locations. So, their spatial locations are not unique, as will be shown in the following experiments. The ground truth identifies only one set of five spatial locations of A, B, C, K, M. The same types of A, B, C, K, M, which are also true endmembers, may also appear at some other locations. In this case, the EIA developed for TE in Section VI-B can be used to identify other spatial locations of these five minerals.

VIII. EXPERIMENTAL RESULTS AND DISCUSSIONS

For real image experiments, the accurate number of true endmembers, n_e is generally unknown and must be either assumed a priori, such as datasets used in [30], [31], [32], [33], [34], and [35], or estimated. As discussed in Section II, VD was used to estimate n_e by n_{VD} for our experiments with $n_{VD} \geq n_e$. In this case, EF is used instead of EE to find VEs. Then, an EF algorithm was used to find n_{VE} VEs, some of which may not be true endmembers. In order to identify true endmembers among these VEs, EIA is designed to accomplish this task.

A. HYDICE Panel Data

1) EF Using Full Bands, BSel, and BSam: According to [16], [17], [18], [20], and [21], VD was estimated to be $n_{VD} = p = 9$. Fig. 13 shows VEs found by ATGP, VCA, and SGA using full bands, two BSel methods, UBS, SMI-BS, and bands sampled by BSam with $n_{VD} = n_{BSel} = n_{BSam} = 9$ and $n_e = 5$, where the found VEs are marked by blue open circles. Since VCA and BSam used random initial conditions, experiments for two random initial condition scenarios indicated by j were also included for comparison. From Fig. 13, none of the results could find panel pixels corresponding to all five true endmembers.

Table VIII tabulates the detected panel pixels as true endmembers. The results clearly showed that SMI-BS produced the best results for ATGP and SGA, which could find four panel pixels in four different rows. On the other hand, RBSam gave the best result for VCA, which could find three panel pixels in three different rows. These best results were still

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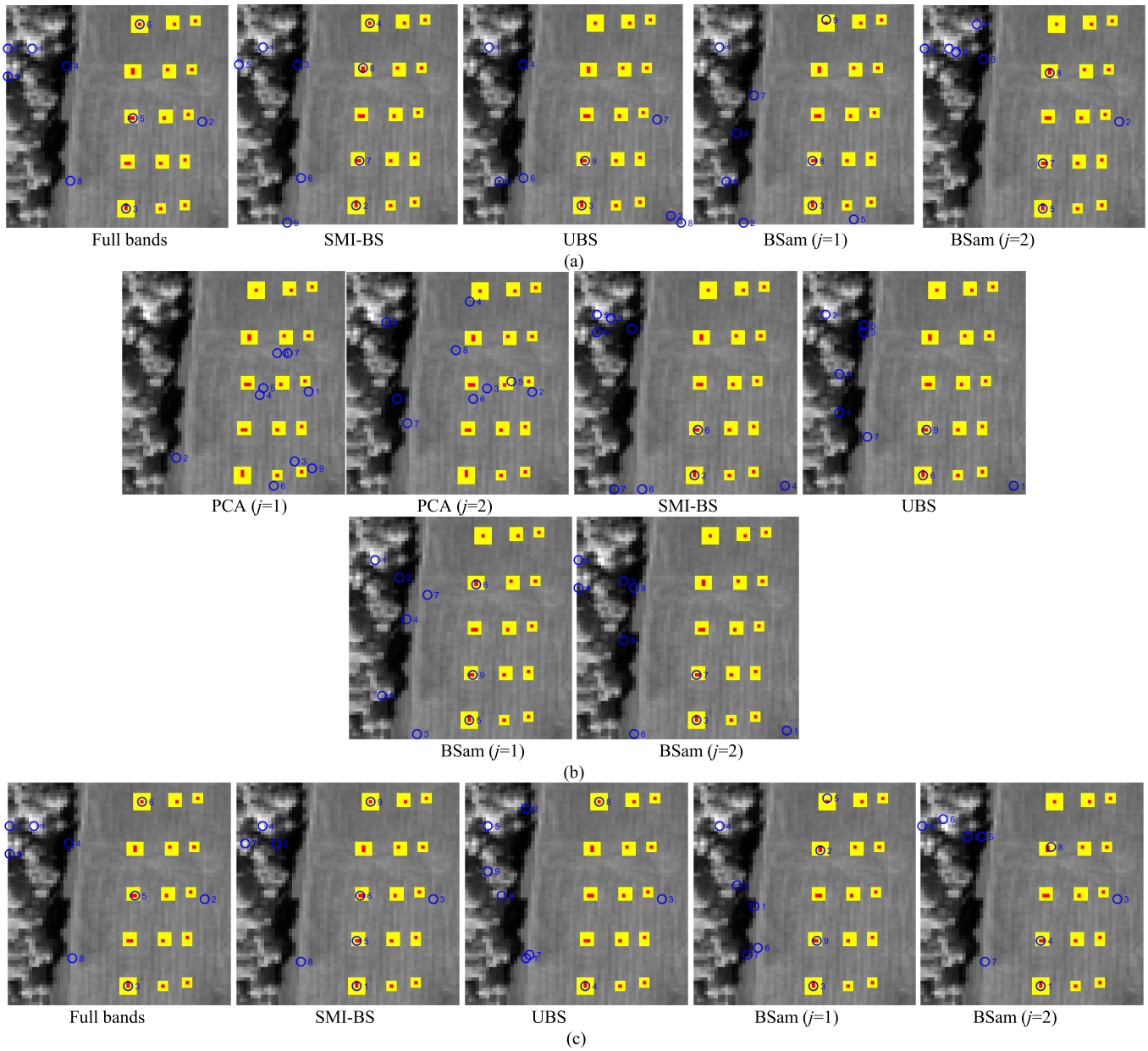


Fig. 13. Endmembers found by (a) ATGP, (b) VCA, and (c) SGA using full bands and bands selected by UBS, SMI-BS, and BSam.

TABLE VIII

ENDMEMBERS FOUND BY ATGP, VCA, AND SGA USING FULL BANDS AND BAND SELECTED BY UBS, SMI-BS, AND RBSAM

	ATGP	VCA	SGA
Full bands	p_{11}, p_{312}, p_{521}	-	p_{11}, p_{312}, p_{521}
PCA	-	- | -	-
SMI-BS	p_{11}, p_{211}, p_{412}, p_{521}	p_{412}, p_{521}	p_{11}, p_{311}, p_{411}, p_{521}
UBS	p_{412}, p_{521}	p_{412}, p_{521}	p_{11}, p_{521}
RBSam	j=1: p_{411}, p_{521} | j=2: p_{221}, p_{311}, p_{521}	j=1: p_{221}, p_{412}, p_{521} | j=2: p_{311}, p_{521}	j=1: p_{221}, p_{312}, p_{521} | j=2: p_{311}, p_{521}

as good as or better than the results obtained by using full bands. Nevertheless, experiments demonstrated that ATGP, VCA, and SGA could not find all five panel pixels as true endmembers, no matter which EF algorithms used full bands or implemented DR or BSel. This indicated that the VD estimated value, $n_{\mathrm{VD}} = p = 9$, is not sufficiently large for EF algorithms to find all desired five panel signatures as true endmembers. Table IX also tabulates their running times.

2) RBSam: As shown in Fig. 13, BSam produced different results due to its randomness caused by RBS. To resolve this issue, RBSam was implemented. Fig. 14 shows the VEs found

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TABLE IX

RUNNING TIME IN SECONDS REQUIRED BY ATGP, VCA, AND SGA USING FULL BANDS AND BAND SELECTED BY UBS, SMI-BS, AND RBSAM

	DR		EFAs			
			ATGP	VCA		SGA
Full band	-		0.0635	-		1.0656
PCA	0.0971		-	0.0417	0.0164	-
SMI-BS	16.9117		0.0038	0.0091		0.1177
Uniform-BS	0.0016		0.0019	0.0133		0.1069
RBSam	0.0204	0.0373	0.0019	0.0030	0.0034	0.1066

TABLE X

ENDMEMBERS FOUND BY ATGP, VCA, AND SGA GENERATED BY RBSAM ($m = 9$, $p = 9$)

	ATGP	VCA	SGA
k =1	p_{11}, p_{211}, p_{412}, p_{521}	p_{11},p_{521}	p_{11}, p_{411}, p_{521}
k =2	p_{11}, p_{211}, p_{311}, p_{412}, p_{521}	p_{11},p_{412}, p_{521}	p_{11}, p_{411}, p_{521}
k =3	-	p_{11}, p_{412}, p_{521}	p_{11}, p_{411}, p_{521}
k =4	-	p_{11}, p_{412}, p_{521}	p_{11}, p_{211},p_{411}, p_{521}
k =5	-	p_{11},p_{212},p_{412}, p_{521}	p_{11}, p_{211},p_{311}, p_{411}, p_{521}
k =6	-	p_{11},p_{212},p_{311}, p_{412}, p_{521}	-

TABLE XI

RUNNING TIMES IN SECONDS REQUIRED BY RBSAM, THREE EF ALGORITHMS, AND RBSAM + EF ALGORITHMS ($m = 9$, $p = 9$)

	RBSam	EF algorithm			RBSam+EF algorithms		
		ATGP	VCA	SGA	RBS-ATGP	RBS-SGA	RBS-VCA
k =1	0.0242	0.0083	0.0492	0.1237	0.0325	0.0734	0.1479
k =2	0.0141	0.0038	0.0086	0.1058	0.0179	0.0227	0.1200
k =3	0.0197	-	0.0047	0.1071	-	0.0245	0.1268
k =4	0.0181	-	0.0148	0.1087	-	0.0329	0.1268
k =5	0.0147	-	0.0088	0.1095	-	0.0235	0.1242
k=6	0.0210	-	0.0029	-	-	0.0239	-
Total time	0.1118	0.0121	0.0890	0.5548	0.0504	0.2008	0.6457

TABLE XII

PANEL PIXELS FOUND BY ATGP, VCA, AND SGA ON PRRBS-SAMPLED BAND SPACE FOR m = 5–9 AND p = 9

Outer loop	Inner loop	ATGP	VCA	SGA
m=5	k=1	p_{11}, p_{411}, p_{521}	p_{521}	p_{411}, p_{521}
	k=2	p_{11}, p_{411}, p_{412}, p_{521}	p_{11}, p_{412}, p_{521}	p_{411}, p_{412}, p_{521}
	k=4	p_{11}, p_{411}, p_{412}, p_{521}	p_{11}, p_{412}, p_{521}	p_{11},p_{411}, p_{412}, p_{521}
	k=7	p_{11},p_{312},p_{411}, p_{412}, p_{521}	p_{11}, p_{312},p_{412}, p_{521}	p_{11},p_{312},p_{411}, p_{412}, p_{521}
	k=10	p_{11},p_{311},p_{312},p_{411}, p_{412}, p_{521}(missing p_2)	p_{11}, p_{312},p_{412}, p_{521} (missing p_2)	p_{11},p_{311},p_{312},p_{411}, p_{412}, p_{521} (missing p_2)
m=6	k=1	p_{11}, p_{412}, p_{521}	p_{11}, p_{411}	p_{412}, p_{521}
	k=2	p_{11}, p_{312},p_{412}, p_{521}	p_{11}, p_{312},p_{411}, p_{521}	p_{11},p_{412}, p_{521}
	k=3	p_{11}, p_{211},p_{312},p_{412}, p_{521}	p_{11},p_{211},p_{312},p_{411}, p_{412},p_{521}	p_{11},p_{211},p_{412}, p_{521}
	k=7	-	-	p_{11},p_{211},p_{312},p_{412}, p_{521}

by RBSam by blue open circles, with k indicating the number of repetitions to run RBSam before a stopping rule is satisfied. By taking unions of VEs obtained for all k repetitions, all three EF algorithms =, ATGO, VCA, and SGA, could all achieve the desired five true endmembers, one R panel pixel for each row.

Table X tabulates the panel pixels found as true endmembers, where the panel pixels highlighted in red indicate that they were found at a particular repetition, k. As we can see, the three EF algorithms using RBSam were able to find all five panel signatures with the least number of repetitions, $k = 2$, required by ATGP, and the largest number of repetitions, $k = 6$, required by VCA. It is worth noting that VCA is a random algorithm using random initial conditions; RBSam was able to fix this random issue through a repetition process. The experiments further showed that VCA did not perform at the same level as ATGP and SGA did. Table XI also tabulates their running times.

3) PRBSam: In RBSam, the BSamR was determined by $n_{\text{VD}} = p = 9$. However, VD is only an estimate, which may be very likely to be overestimated. PBSam provides this evidence to show that it is indeed the case. To save space, the figures showing the found endmembers are not included. Instead, Table XII tabulates the panel pixels found as endmembers

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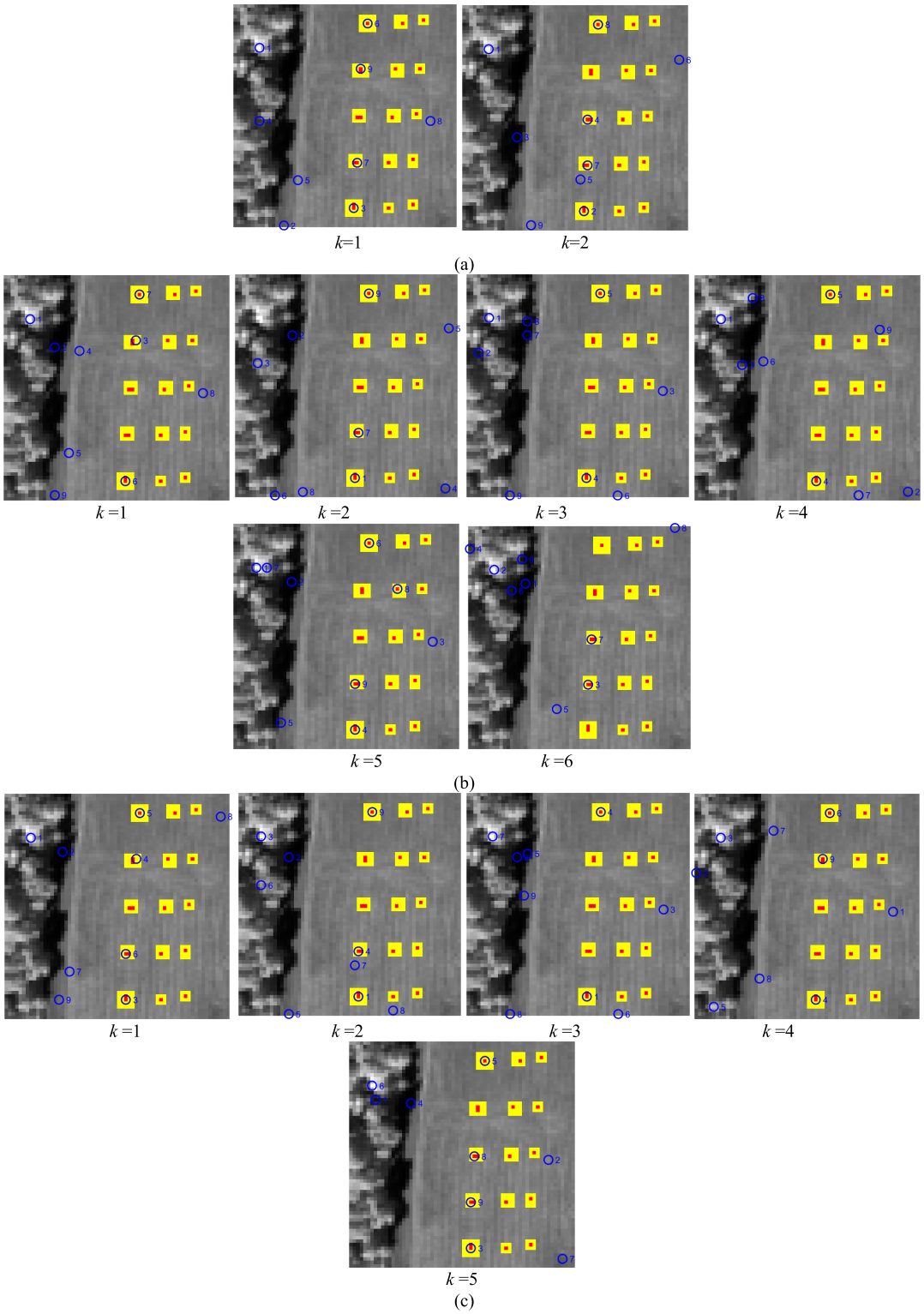


Fig. 14. Endmembers found by (a) ATGP, (b) VCA, and (c) SGA using full bands and bands selected by UBS, SMI-BS, and RBSam.

found by ATGP, VCA, and SGA on PRBS-sampled band space for $m = 5$–9 and $p = 9$, where the panel pixel highlighted in red indicates that it was found at a particular repetition, k, for a given m.

Upon implementing PBSam, the index $n_{\text{BSam}} = m$ is used to keep track of the number of compressively sensed bands, and the index k is used to dictate the number of repetitions required for each m before it moves to the next progression $m+1$. It clearly shows from Table XII that it only needed six compressively sensed bands, i.e., $m = n_{\text{BSam}} = 6$, to find all the five panel pixels corresponding to five different rows as five distinct endmembers. However, according to Table VIII,

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TABLE XIII
SAM VALUES BETWEEN EIA-IDENTIFIED ENDMEMBERS AND THEIR CORRESPONDING TRUE ENDMEMBERS WITH SAM $< \varepsilon = 0.05$

		A	B	C	K	M
ATGP	Full bands	**0.0167**	**0.0334**	0.0516	0.0613	**0.0000**
	SMI-BS	0.0921	0.0546	**0.0474**	**0.0300**	**0.0229**
	UBS	0.0630	0.0671	0.0629	**0.0220**	**0.0264**
	BSam(j=1)	**0.0263**	**0.0218**	**0.0454**	**0.0000**	0.0651
	BSam(j=2)	**0.0472**	**0.0380**	0.0725	**0.0220**	0.0662
VCA	PCA(j=1)	**0.0411**	0.0650	**0.0459**	0.0695	0.0600
	PCA(j=2)	0.0529	0.0583	**0.0449**	0.0725	0.0663
	SMI-BS	**0.0350**	0.0649	0.0643	0.0760	**0.0343**
	UBS	0.0943	0.0691	0.0611	0.1110	0.0615
	BSam(j=1)	0.0774	0.0881	0.0688	**0.0342**	0.0692
	BSam(j=2)	**0.0172**	0.0556	**0.0444**	**0.0220**	**0.0249**
SGA	Full bands	**0.0167**	**0.0334**	0.0516	0.0613	**0.0000**
	SMI-BS	0.0732	**0.0218**	0.0516	0.0611	**0.0264**
	Uniform-BS	**0.0172**	0.0780	0.0692	**0.0348**	**0.0000**
	BSam(j=1)	0.1101	0.0622	0.0725	0.0899	0.0557
	BSam(j=2)	**0.0490**	0.0720	0.0611	**0.0348**	**0.0000**

the value of 9 was not sufficient for BSam without repetitions to find all five endmembers. These experiments showed that the value of 9 estimated by VD for RBSam was indeed overestimated.

B. Cuprite Data

The Cuprite scene is very different from the HYDICE panel data in Fig. 12 in the sense that the true endmembers of R panel pixels in HYDICE data can be identified by their spatial locations, while the spatial locations of minerals in the Cuprite data in Fig. 4 can be only identified by their spectral signatures. In other words, the R panel pixels as true endmembers in Fig. 12 can be identified by their unique spatial locations according to the ground truth. By contrast, the spatial locations of the minerals as endmembers in Fig. 4 are not unique and may appear in different locations. As a consequence, the spatial locations of these minerals in Fig. 4(b) cannot be used to identify the true endmembers because the same type of mineral signature may appear in various spatial locations. In this case, EIA was used to identify the desired true minerals among VEs found by a given EF algorithm. The VD estimated for the Cuprite scene by the HFC method with $P_F \leq 10^{-2}$ was $n_{VD} = p = 22$ [17], [18], [20], [21].

A comment on EIA is noteworthy. Since there is no prior knowledge of n_e, the VD-estimated value, $n_{VD} = p$, is assumed to be the number of VEs as spectrally distinct signatures. So, p is generally no less than n_e, i.e., $n_{VE} = p \geq n_e$, and EIA is supposed to find n_e true endmembers from the p VEs found by an EF algorithm. Two scenarios often occur with $n_{VE} = p \geq n_e$.

1) If the error threshold, ε imposed on SAM values in step 3 of EIA, is set too large, EIA may mistakenly find VEs for correct true endmembers.
2) If the error threshold, ε imposed on SAM values in step 3 of EIA, is set too small, EIA may not be able to find all n_e correct true endmembers.

The value of $\varepsilon = 0.05$ was chosen empirically and found to be effective in our experiments.

1) 1RBSam: Fig. 15(a)–(c) shows spatial locations of the five ground truth minerals, A, B, C, K, M marked by red circles and the five VEs marked by blue open circles which were identified by EIA with SAM $< \varepsilon = 0.05$ as true endmembers from 22 VEs found by the three EF algorithms, ATGP, VCA, and SGA using full bands, bands selected by UBS, SMI-BS, and BSam with $m = 22$. Since the bands sampled by BSam were random, two scenarios were included with $j = 1$ and 2.

For quantitative analysis, Table XIII tabulates SAM values between EIA-identified blue colored endmembers and their corresponding red-colored true endmembers with SAM <0.05, where the identified VEs as true endmembers are boldfaced.

As shown in Fig. 15(a)–(c) and Table XIII, only three out of 22 VEs could be identified by EIA as three different mineral endmembers. Also, except for a few EIA-identified red-colored mineral M, which were exactly the same ground truth endmembers, all other blue-colored minerals identified by EIA had different spatial locations from those of the ground truth endmembers. More importantly, the VEs found by each EF algorithm were actually different and occurred at different spatial locations. These experiments also demonstrated that ATGP, VCA, and SGA could not find all five minerals, either using full bands, BSel, or BSam. Also, these experiments showed that when the same endmembers are present in multiple locations in the data scene, true endmembers cannot be simply identified by their spatial locations but rather by their spectral signatures.

Fig. 16 compares the similarity of spectral profiles between EIA-found endmembers in Table XIII and the ground truth endmembers for comparison. Table XIV tabulates the running times required by three EF algorithms, ATGP, VCA, and SGA, using the bands selected by various BS methods, UBS, SMI-BS, and BSam.

2) 2RBSam: Since BSam produced different results, as shown in Fig. 15, RBSam was implemented to fix the randomness issue. Fig. 17 shows the spatial locations of the five ground truth minerals, A, B, C, K, M, marked as red circles and the five VEs marked as blue open circles, which

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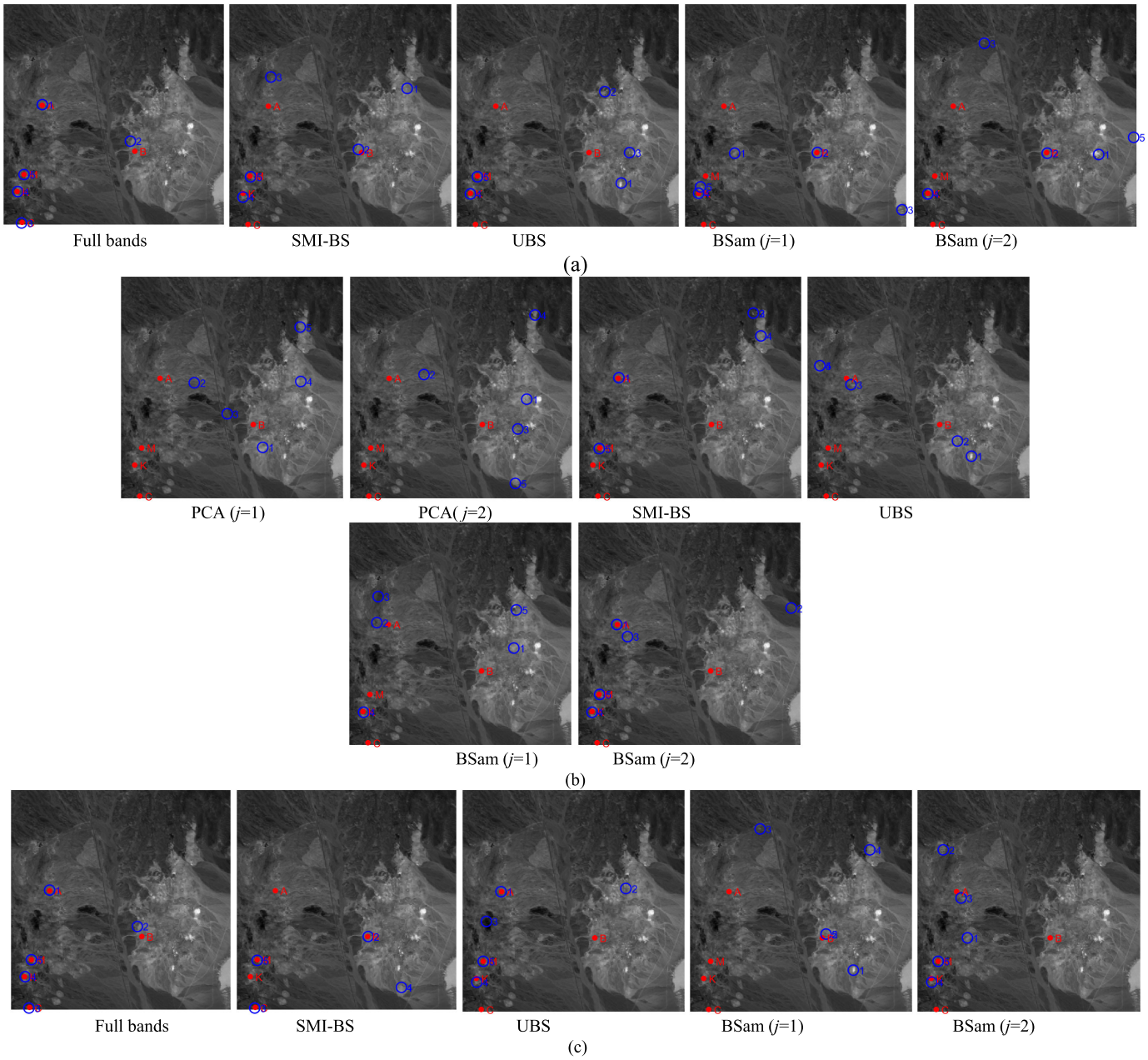


Fig. 15. Endmembers found by (a) ATGP, (b) VCA, and (c) SGA using full bands and bands selected by UBS, SMI-BS, and BSam.

TABLE XIV

RUNNING TIME IN SECONDS THREE EF ALGORITHMS ATGP, VCA, SGA USING FULL BANDS AND BANDS SELECTED BY V UBS, SMI-BS, RBSAM

	DR	EFAs			DR+EFAs		
		ATGP	VCA	SGA	ATGP	VCA	SGA
ODS	-	1.5340	-	205.6901	1.6591	-	205.8153
PCA(j=1)	0.1252	-	0.0871	-	-	0.2123	-
PCA(j=2)	0.1177	-	0.0557	-	-	0.1733	-
SMI-BS	69.1149	0.1353	0.0527	10.5439	0.2529	69.1677	10.6616
Uniform-BS	0.0012	0.1324	0.0546	10.4405	69.2473	0.0558	79.5554
RBSam(j=1)	1.3319	0.1295	0.0670	10.4482	0.1307	1.3989	10.4493
RBSam(j=2)	1.3966	0.1356	0.0704	10.4874	1.4675	1.4670	11.8192

were identified by EIA with SAM $< \varepsilon = 0.05$ to identify five blue colored endmembers from 22 VEs found by the three EF algorithms, ATGP, VCA, and SGA, using full bands, bands selected by UBS, SMI-BS, and RBSam with $m = 22$ and k indicating how many repetitions required by each EF algorithm. As we can see from Fig. 17, none of EIA identified

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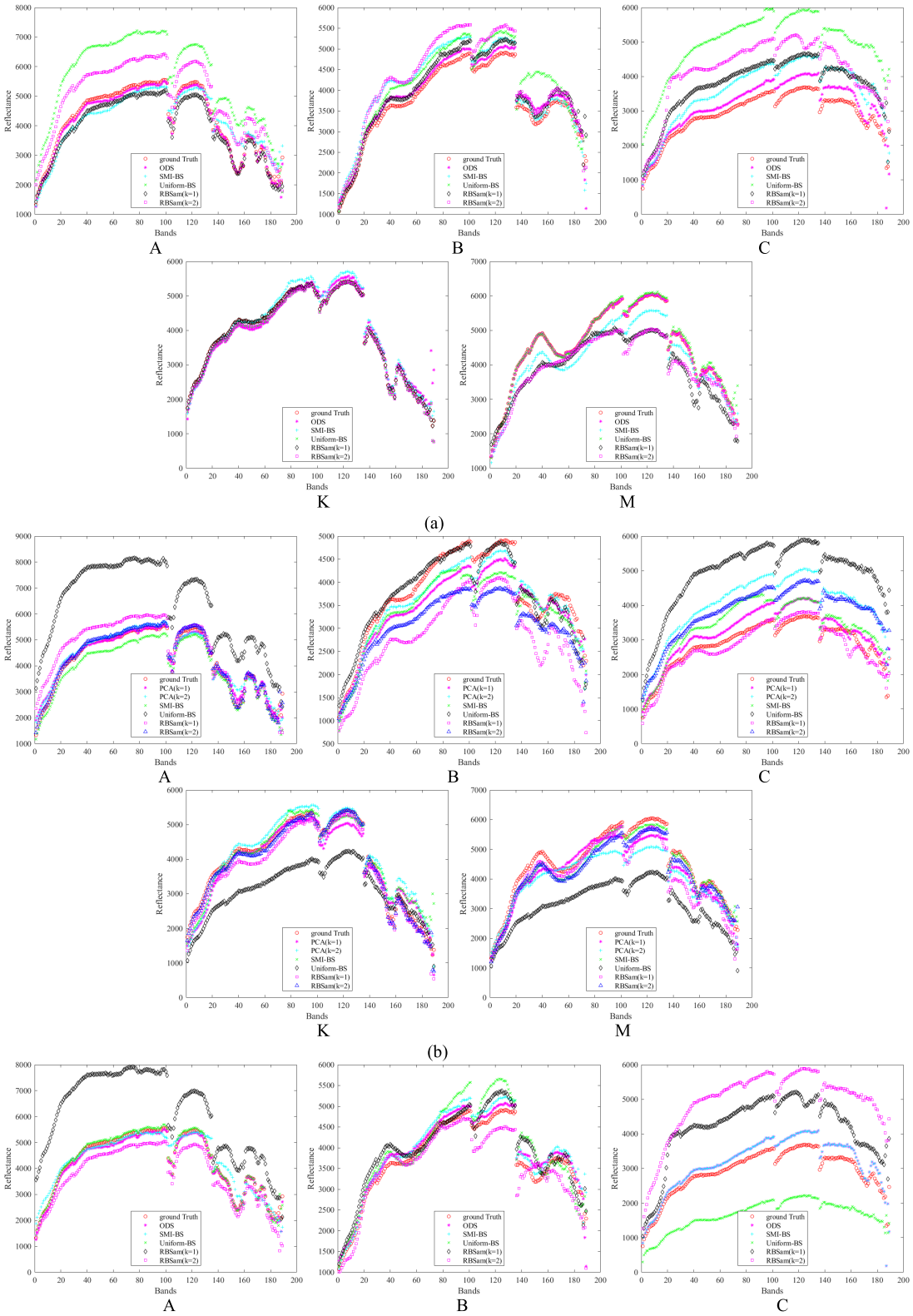
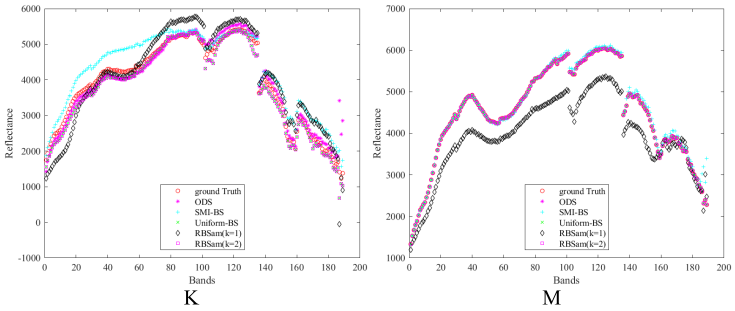


Fig. 16. Similarity of spectral profiles between EIA-found endmembers and ground truth endmembers. (a) ATGP. (b) VCA. (c) SGA.

VEs were identical to the same red-colored ground truth endmembers. Nevertheless, the three EF algorithms could find all five true mineral endmembers.

For quantitative analysis, Table XV tabulates SAM values between EIA-identified blue colored endmembers and their corresponding red-colored true endmembers with SAM <0.05, where the identified VEs as true endmembers are boldfaced. As we can see from Table XV, all three EF algorithms using RBSam were able to find all five true mineral endmembers with the least number of repetitions, $k = 2$, required by

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(c)

Fig. 16. *(Continued.)*Similarity of spectral profiles between EIA-found endmembers and ground truth endmembers. (a) ATGP. (b) VCA. (c) SGA.

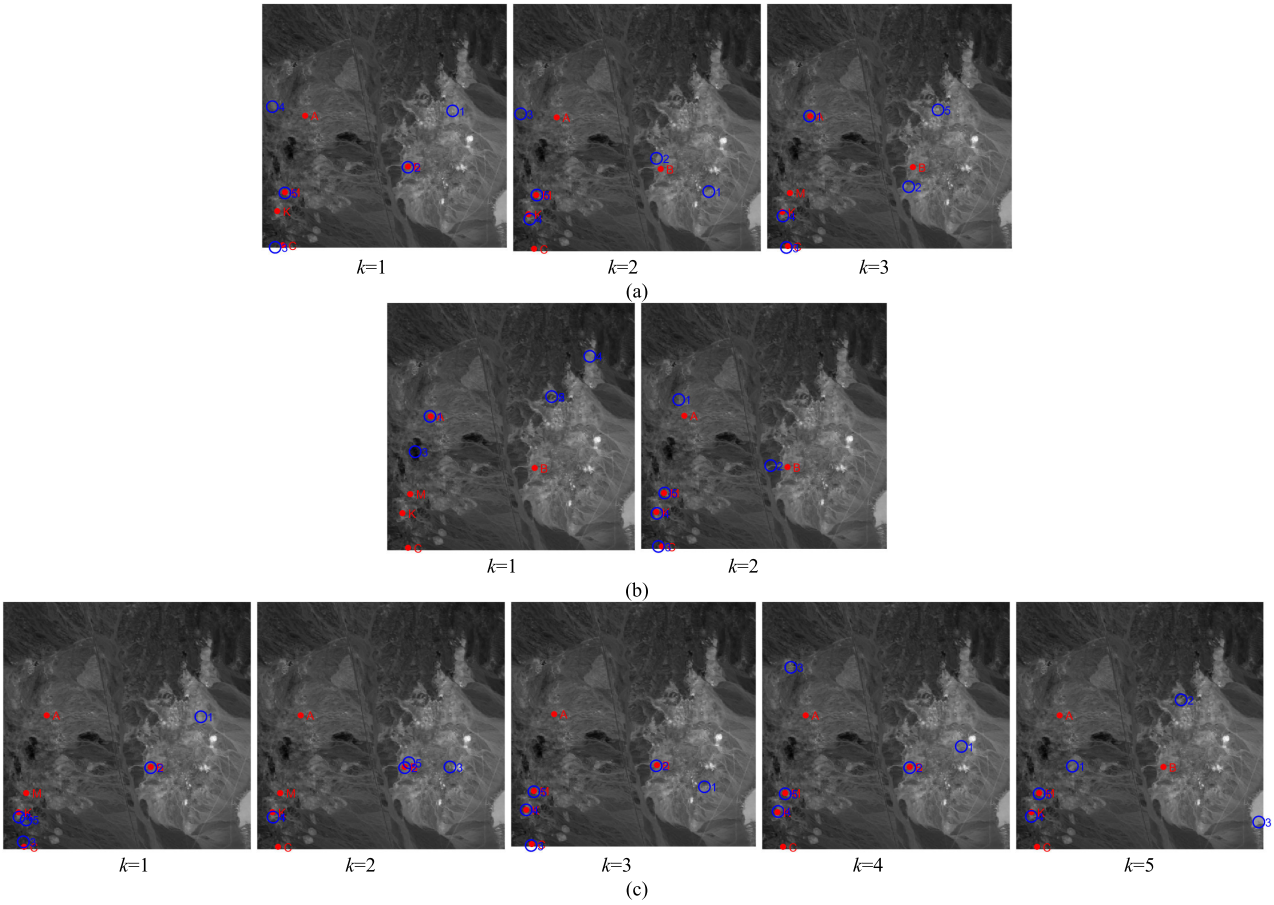


Fig. 17. Endmembers found by ATGP, VCA, and SGA using full bands and bands selected by UBS, SMI-BS, and RBSam. (a) ATGP. (b) VCA. (c) SGA.

VCA, and the largest number of repetitions, $k = 5$, required by SGA. Fig. 18 compares the similarity of spectral profiles between EIA-found endmembers in Table XIII and the ground truth endmembers for comparison. Table XVI also tabulates their running times. EIA-found VEs and ground truth endmembers.

3) 3PRBSam: Apparently, compared to $n_e = 5$, $p = 22$ was significantly overestimated. So, PBSam was implemented to see how low p can be. In this case, we started with half of VD, i.e., $p = 11$, in which case, m was set to 11. Then, ran RBSam progressively one at a time until it found all five minerals. Since too many figures are required to show spatial locations of all VEs found by three EF algorithms, to save space only the VEs found by ATGP, VCA and SGA using EIA via SAM $< \varepsilon = 0.05$ are tabulated in Tables XVII–XIX, respectively, with m starting 11 and gradually increasing by one until it found all the five minerals where the VEs identified as true endmembers are boldfaced. In the meantime, RBSam ran for each m with k indicating the number of repetitions. According to Tables XVI–XVIII, ATGP, VCA, and SGA could find all five minerals for $m = 12$ with $k = 10$, $m = 12$ with $k = 10$, and $m = 11$ with $k = 9$, respectively, and SGA performed the best. Apparently, $m = 22$ was set too large for RBSam.

C. Discussions

The experiments conducted for the HYDICE panel data in Section VII-A and the Cuprite data in VII.B reveal several interesting findings.

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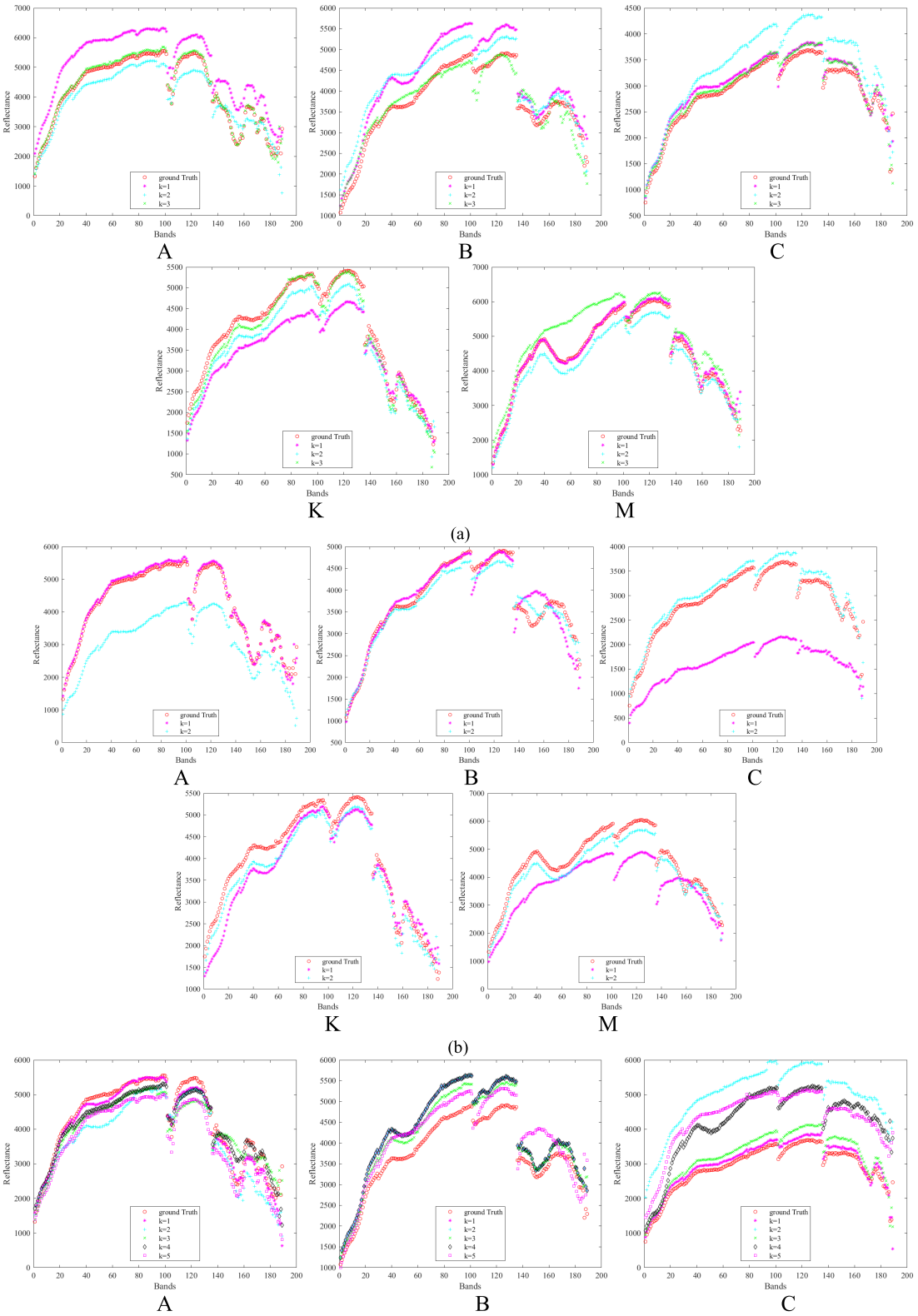
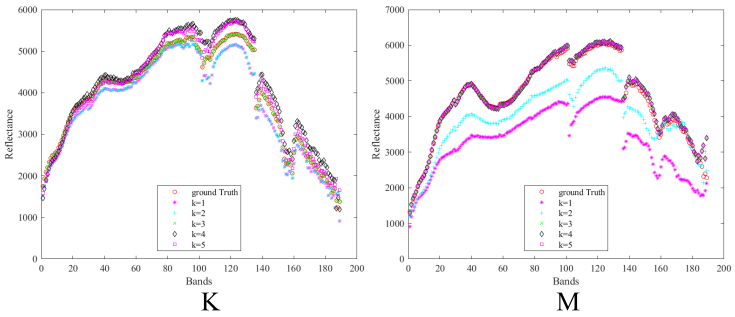


Fig. 18. Similarity of spectral profiles between EIA-found endmembers and ground truth endmembers. (a) ATGP. (b) VCA. (c) SGA.

1) Although EF does not require DR or BSel to reduce spectral dimensionality, EF using RBSam can significantly improve the EF performance. In particular, the BSam rate, n_{BSam}, can be much smaller than the VD-estimated value, n_{VD}, such as the Cuprite data.

2) Since EF is unsupervised, the VEs found by an EF algorithm are generally unknown. In this case, the EF algorithm-found VEs are considered as potential endmember candidates. Whether or not VEs are indeed true endmembers must be validated by ground truth.

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(c)

Fig. 18. *(Continued.)*Similarity of spectral profiles between EIA-found endmembers and ground truth endmembers. (a) ATGP. (b) VCA. (c) SGA.

TABLE XV

FIVE ENDMEMBERS IDENTIFIED BY EIA USING $\varepsilon = 0.05$ AS AN SAM THRESHOLD VALUE AMONG 22 FOUND IN FIG. 12

		A	B	C	K	M
ATGP	k =1	0.0565	**0.0383**	**0.0379**	0.0625	**0.0264**
	k =2	0.0704	-	-	**0.0314**	-
	k =3	**0.0172**	-	-	-	-
VCA	k =1	**0.0172**	0.0730	**0.0474**	0.0808	0.0899
	k =2	-	**0.0405**	-	**0.0342**	**0.0249**
SGA	k =1	0.0607	**0.0383**	0.0519	**0.0341**	0.0595
	k =2	0.0889	-	0.0629	-	0.0557
	k =3	0.0830	-	**0.0398**	-	**0.0264**
	k =4	0.0669	-	-	-	-
	k=5	**0.0489**	-	-	-	-

TABLE XVI

RUNNING TIME IN SECONDS THREE EF ALGORITHMS ATGP, VCA, SGA USING FULL BANDS AND BANDS SELECTED BY UBS, SMI-BS, RBSAM WITH $m = 22$

	RBS	ATGP	VCA	SGA
k =1	1.4714	0.3281	1.0495	10.7261
k =2	1.3457	0.1761	0.0781	10.5056
k =3	1.3421	0.1396	-	10.5245
k =4	1.2771	-	-	10.5274
k=5	1.3606	-	-	10.5386

3) EF using RBSam was shown to perform better than EF using full bands, DR, or BSel because it repetitively implements RBSam and integrates their endmember results by taking the unions of VEs produced by RBSam using all repetitions.
4) The experimental results showed that PRBSam worked much better than RBSam and could actually accomplish EF with a lower value of VD-estimated p.
5) As shown in the experiments, the VD-estimated value, $n_{\mathrm{VD}} = p$, is generally greater than the true n_{e}. On some occasions, such as TI and TE, $n_{\mathrm{VE}} = 6$ is very close to $n_{\mathrm{e}} = 5$, and HYDICE data, $n_{\mathrm{VE}} = 9$ is also close to $n_{\mathrm{e}} = 5$. On another occasion, the Cuprite data $n_{\mathrm{VD}} = 22$ is much greater than $n_{\mathrm{e}} = 5$. In order to address this issue, PRBSam was proposed in this article, which can start with a lower value of p, denoted by m, and then progressively increase m by one at a time. Experiments demonstrated that the BSam rate, n_{BSam}, can actually be much lower than the n_{VD} value.
6) As also shown in the experiments, BSam is a dynamic BSel as opposed to UBS and SMI-BS, which are static BSel methods. So, a dynamic BSel can do better than what a static BSel algorithm does, as shown in [29].
7) Last but not least, experiments performed in this article also demonstrated an important fact that datasets used for EF were crucial. The HYDICE panel data and Cuprite data exhibit complete characteristics of image scenes. For example, HYDICE data clearly show the exact spatial locations of pure panel signatures as endmembers. In this case, endmembers found by an EF algorithm can be verified and validated by their spatial locations. On the other hand, the Cuprite has many same pure mineral signatures located in different locations. The EF-found endmembers cannot be identified simply by their spatial locations but rather by their spectral signatures. This is the main reason that EIA was developed for this purpose. Unfortunately, the datasets used by most EF works reported in the literature belong to this category, such as Houston, Pavia, Jasper

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TABLE XVII
ENDMEMBERS FOUND BY ATGP USING PRBBSAM AND IDENTIFIED BY EIA WITH SAM VALES <0. 05

	m	k	A	B	C	K	M	Found endmembers
ATGP	m=11	k=1	0.0822	0.0773	0.0701	**0.0220**	0.0860	K
		k=2	**0.0490**	0.0730	**0.0280**	0.0926	**0.0264**	A,C,K,M
		k=3	0.0490	0.0685	0.0725	0.0300	0.0501	A,C,K,M
		k=4	0.1076	0.0745	0.0966	0.0220	0.0577	A,C,K,M
		k=5	0.0226	0.0802	0.0725	0.0613	0.0974	A,C,K,M
		k=6	0.0172	0.0781	0.0725	0.0431	0.0706	A,C,K,M
		k=7	0.0889	0.0777	0.0725	0.1354	0.0830	A,C,K,M
		k=8	0.0490	0.0641	0.0558	0.0660	0.0264	A,C,K,M
		k=9	0.0235	0.1028	0.0362	0.0220	0.1093	A,C,K,M
		k=10	0.1098	0.0805	0.0511	0.0613	0.0595	A,C,K,M (missing B)
	m=12	k=1	0.1076	0.0520	0.0665	**0.0220**	**0.0000**	K,M
		k=2	0.1053	0.0782	0.0725	0.0342	0.0264	K,M
		k=3	0.1008	0.0782	0.0746	0.0613	0.0692	K,M
		k=4	**0.0000**	**0.0281**	0.0723	0.0220	0.0264	A,B,K,M
		k=5	0.1101	0.0749	0.0701	0.0342	0.0264	A,B,K,M
		k=6	0.1044	0.0688	0.0928	0.0342	0.0809	A,B,K,M
		k=7	0.0000	0.0782	0.0629	0.0300	0.0264	A,B,K,M
		k=8	0.0000	0.0781	0.0528	0.0954	0.0264	A,B,K,M
		k=9	0.1030	0.0383	0.0982	0.0300	0.0000	A,B,K,M
		k=10	0.1098	0.0804	**0.0350**	0.0613	0.0883	A,B,C,K,M

TABLE XVIII
ENDMEMBERS FOUND BY VCA USING PRBBSAM AND IDENTIFIED BY EIA WITH SAM VALES <0.05

	m	k	A	B	C	K	M	Found endmembers
VCA	m=11	k=1	0.0771	0.0726	0.1035	**0.0342**	0.1013	K
		k=2	0.0958	**0.0281**	**0.0350**	-	0.0684	B,C,K
		k=3-10	0.0606	-	-	0.0342	0.0687	B,C,K (missing A,M)
	m=12	k=1	0.1177	0.0869	0.0725	**0.0342**	0.0974	K
		k=2	0.0674	0.0918	0.0725	-	0.0974	K
		k=3	0.1027	**0.0281**	0.1079	-	0.0968	B,K
		k=4	0.0912	-	0.0688	-	0.0974	B,K
		k=5	**0.0235**	-	0.0876	-	0.0749	A,B,K
		k=6	-	-	0.0954	-	**0.0483**	A,B,K,M
		k=7	-	-	0.0972	-	-	A,B,K,M
		k=8	-	-	0.0677	-	-	A,B,K,M
		k=9	-	-	0.0851	-	-	A,B,K,M
		k=10	-	-	**0.0350**	-	-	A,B,C,K,M

TABLE XIX
ENDMEMBERS FOUND BY SGA USING PRBBSAM AND IDENTIFIED BY EIA WITH SAM VALES <0.05

	m	k	A	B	C	K	M	Found endmembers
SGA	m=11	k=1	0.1025	0.0885	0.0594	0.0613	**0.0264**	M
		k=2	0.1101	0.0641	0.0558	0.1413	0.0264	M
		k=3	0.0626	**0.0383**	0.0835	**0.0220**	0.0264	B,K,M
		k=4	0.1027	0.0804	0.0960	0.0220	0.0749	B,K,M
		k=5	**0.0172**	0.0780	0.0725	0.0220	0.0850	A,B,K,M
		k=6	0.0310	0.0932	0.0725	0.0314	0.0264	A,B,K,M
		k=7	0.1101	0.0777	0.0960	0.1282	0.0822	A,B,K,M
		k=8	0.0558	0.0671	0.0655	0.0671	0.0264	A,B,K,M
		k=9	0.0672	0.1028	**0.0362**	0.0687	0.0905	A,B,C,K,M

Ridge datasets, etc., in which case the endmembers are basically determined by human visual inspection. Technically speaking, this practice is not valid.

IX. CONCLUSION

BSam, originally coined and proposed by Chang [11], has found its applications in hyperspectral target and anomaly detection and hyperspectral image classification. However, its applicability to EF is yet justified. This is mainly due to the fact that its used iterative feedback process to capture spatial information smears and contaminates the signature purity of found endmembers. To resolve this issue, this article revisits BSam to further rederive a repetitive RBSam, which replaces the iterative feedback process with a repetitive process

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without using spatial filters. It turns out that RBSam is much simpler than BSam and easy to implement. Most importantly, it also works very effectively for EF. In order to deal with an inaccurate estimate of n_e, a progressive version of RBSam, PRBSam, is further developed, which demonstrated that n_{RSam} can actually be lower than the VD-estimated value, $n_{\text{VD}} = p$. This significant advantage results from RBSam using the concept of RBS replacing RSS in CS but cannot be gained from any other static DR technique.

As a final concluding remark, it is worth noting that the BSam rate can be lower than the actual number of endmembers, as shown by the experiments of PRBSam. Addressing this issue is well beyond the scope of this article because it will require extensive background in information theory using the concept of mutual information [36], channel capacity [37], and rate distortion function [38] to fully explore this issue. Nevertheless, efforts on investigation among BSel, BSam, and compressive band sensing in various applications such as detection, classification, SU, and EF are currently undertaken, which will result in new open research directions. In particular, with the use of the universality model in [39], there is no need for signal recovery for HSI, in which the random projection used for signal recovery is no longer necessary, as in [40].

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Chein-I Chang (Life Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He is currently a Professor with the University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA. He has been a Chang Jiang Scholar Chair Professor and the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, since 2016, and is also a Yushan Scholar Chair Professor at National Cheng Kung University, Tainan, Taiwan. In addition, he has been the Chair Professor of National Yang Ming Chiao Tung University, Hsinchu, Taiwan, since 2019. He has authored five books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real-Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection* (Springer, 2016), *Real-Time Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017), and *Statistical Hyperspectral Target and Anomaly Detection* (Wiley, 2026); and also edited three books *Recent Advances in Hyperspectral Signal and Image Processing* in 2006, *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007), and *Advances in Hyperspectral Image Processing Techniques* (Wiley, 2023). In addition, he also co-edited with A. Plaza wrote a book on *High Performance Computing in Remote Sensing* (CRC Press) in 2007. His research interests include multispectral/hyperspectral image processing and medical imaging.

Dr. Chang is a fellow of SPIE and is serving as Associate Editor for *Remote Sensing* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Lulu Liu received the B.E. degree in software engineering from Henan University of Technology, Zhengzhou, China, in 2021. She is currently pursuing the Ph.D. degree in computer science and technology with Dalian Maritime University, Dalian, China.

Her research interests include hyperspectral compression, hyperspectral sampling, and applications.



Meiping Song received the Ph.D. degree from the College of Computer Science and Technology, Harbin Engineering University, Harbin, China, in 2006.

From 2013 to 2014, she was a Visiting Associate Research Scholar with the Remote Sensing Signal and Image Processing Laboratory, University of Maryland, Baltimore County, Baltimore, MD, USA. She has been a Professor with the College of Information Science and Technology, Dalian Maritime University, Dalian, China, since 2020. Her research interests include remote sensing and hyperspectral image processing.



Chunyan Yu (Senior Member, IEEE) received the Ph.D. degree in environmental engineering from Dalian Maritime University, Dalian, China, in 2012.

She is currently a Professor with the Information Science and Technology College, Dalian Maritime University. Her research interests include remote sensing image processing and pattern recognition.



Tingting Yang received the B.S. and M.S. degrees in communication engineering and computer technology from Liaoning Shihua University, Fushun, China, in 2016 and 2020, respectively, and the Ph.D. degree in information and communication engineering from Dalian Maritime University, Dalian, China, in 2024.

She is currently with the College of Computer Science and Artificial Intelligence, Bohai University, Jinzhou, China. Her research includes hyperspectral image processing.



Hongju Cao received the Ph.D. degree in technology of computer applications from Dalian Maritime University, Dalian, China, in 2022.

He is currently an Associate Professor with Dalian University of Foreign Languages, Dalian. His research interests include anomaly detection and classification in hyperspectral image processing.

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